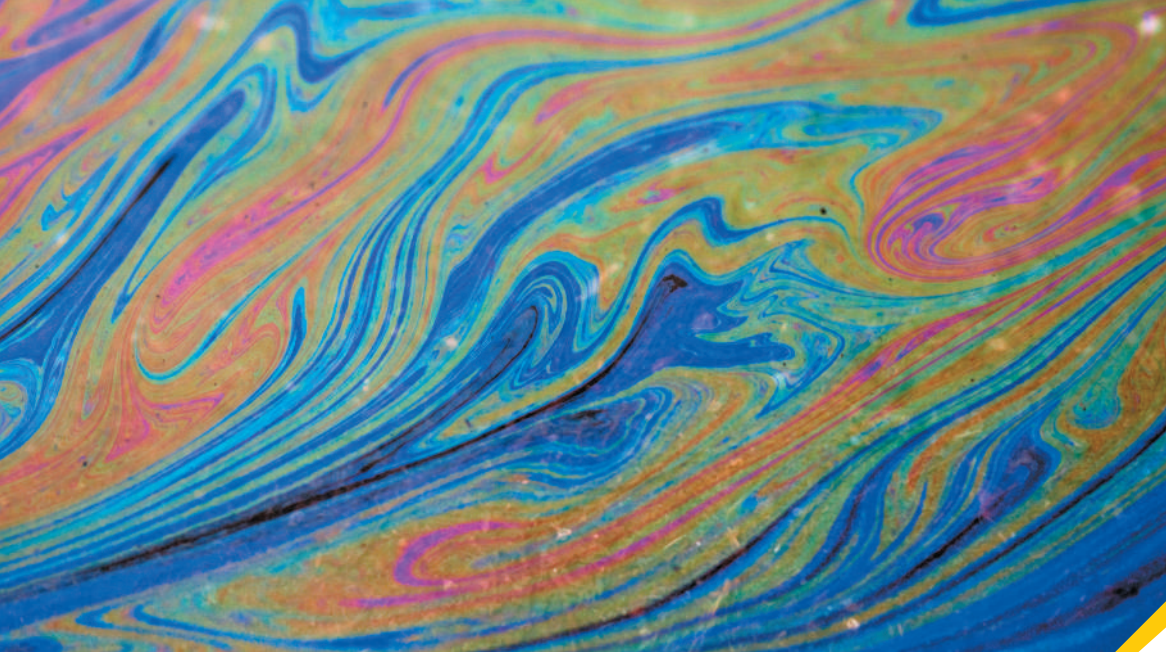




? When white light shines downward on a thin, horizontal layer of oil, light waves reflected from the upper and lower surfaces of the film of oil interfere, producing vibrant colors. The color that you see reflected from a certain spot on the film depends on (i) the film thickness at that spot; (ii) the index of refraction of the oil; (iii) the index of refraction of the material below the oil; (iv) both (i) and (ii); (v) all of (i), (ii), and (iii).

35 Interference

An ugly black oil spot on the pavement can become a thing of beauty after a rain, when the oil reflects a rainbow of colors. Multicolored reflections can also be seen from the surfaces of soap bubbles and DVDs. How is it possible for colorless objects to produce these remarkable colors?

In our discussion of lenses, mirrors, and optical instruments we used the model of *geometric optics*, in which we represent light as *rays*, straight lines that are bent at a reflecting or refracting surface. But light is fundamentally a *wave*, and in some situations we have to consider its wave properties explicitly. If two or more light waves of the same frequency overlap at a point, the total effect depends on the *phases* of the waves as well as their amplitudes. The resulting patterns of light are a result of the *wave* nature of light and cannot be understood on the basis of rays. Optical effects that depend on the wave nature of light are grouped under the heading **physical optics**.

In this chapter we'll look at *interference* phenomena that occur when two waves combine. Effects that occur when *many* sources of waves are present are called *diffraction* phenomena; we'll study these in Chapter 36. In that chapter we'll see that diffraction effects occur whenever a wave passes through an aperture or around an obstacle. They are important in practical applications of physical optics such as diffraction gratings, x-ray diffraction, and holography.

While our primary concern is with light, interference and diffraction can occur with waves of *any* kind. As we go along, we'll point out applications to other types of waves such as sound and water waves.

35.1 INTERFERENCE AND COHERENT SOURCES

As we discussed in Chapter 15, the term **interference** refers to any situation in which two or more waves overlap in space. When this occurs, the total wave at any point at any instant of time is governed by the **principle of superposition**, which we introduced in

LEARNING OUTCOMES

In this chapter, you'll learn...

- 35.1** What happens when two waves combine, or interfere, in space.
- 35.2** How to understand the interference pattern formed by the interference of two coherent light waves.
- 35.3** How to calculate the intensity at various points in an interference pattern.
- 35.4** How interference occurs when light reflects from the two surfaces of a thin film.
- 35.5** How interference makes it possible to measure extremely small distances.

You'll need to review...

- 14.2, 31.1** Phasors.
- 15.3, 15.6, 15.7** Wave number, wave superposition, standing waves on a string.
- 16.4** Standing sound waves.
- 32.1, 32.4, 32.5** Electromagnetic spectrum, wave intensity, standing electromagnetic waves.

Section 15.6 in the context of waves on a string. This principle also applies to electromagnetic waves and is the most important principle in all of physical optics.

PRINCIPLE OF SUPERPOSITION When two or more waves overlap, the resultant displacement at any point and at any instant is found by adding the instantaneous displacements that would be produced at the point by the individual waves if each were present alone.

(In some special situations, such as electromagnetic waves propagating in a crystal, this principle may not apply. A discussion of these situations is beyond our scope.)

We use the term “displacement” in a general sense. With waves on the surface of a liquid, we mean the actual displacement of the surface above or below its normal level. With sound waves, the term refers to the excess or deficiency of pressure. For electromagnetic waves, we usually mean a specific component of electric or magnetic field.

Interference in Two or Three Dimensions

We have already discussed one important case of interference, in which two identical waves propagating in opposite directions combine to produce a *standing wave*. We saw this in Section 15.7 for transverse waves on a string and in Section 16.4 for longitudinal waves in a fluid filling a pipe; we described the same phenomenon for electromagnetic waves in Section 32.5. In all of these cases the waves propagated along only a single axis: along a string, along the length of a fluid-filled pipe, or along the propagation direction of an electromagnetic plane wave. But light waves can (and do) travel in *two* or *three* dimensions, as can any kind of wave that propagates in a two- or three-dimensional medium. In this section we’ll see what happens when we combine waves that spread out in two or three dimensions from a pair of identical wave sources.

Interference effects are most easily seen when we combine *sinusoidal* waves with a single frequency f and wavelength λ . **Figure 35.1** shows a “snapshot” of a *single* source S_1 of sinusoidal waves and some of the wave fronts produced by this source. The figure shows only the wave fronts corresponding to wave *crests*, so the spacing between successive wave fronts is one wavelength. The material surrounding S_1 is uniform, so the wave speed is the same in all directions, and there is no refraction (and hence no bending of the wave fronts). If the waves are two-dimensional, like waves on the surface of a liquid, the circles in Fig. 35.1 represent circular wave fronts; if the waves propagate in three dimensions, the circles represent spherical wave fronts spreading away from S_1 .

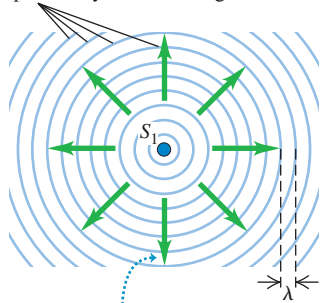
In optics, sinusoidal waves are characteristic of **monochromatic light** (light of a single color). While it’s fairly easy to make water waves or sound waves of a single frequency, common sources of light *do not* emit monochromatic (single-frequency) light. For example, incandescent light bulbs and flames emit a continuous distribution of wavelengths. By far the most nearly monochromatic light source is the *laser*. An example is the helium–neon laser, which emits red light at 632.8 nm with a wavelength range of the order of ± 0.000001 nm, or about one part in 10^9 . In this chapter and the next, we’ll assume that we are working with monochromatic waves (unless we explicitly state otherwise).

Constructive and Destructive Interference

Figure 35.2a shows two identical sources of monochromatic waves, S_1 and S_2 . The two sources produce waves of the same amplitude and the same wavelength λ . In addition, the two sources are permanently *in phase*; they vibrate in unison. They might be two loudspeakers driven by the same amplifier, two radio antennas powered by the same transmitter, or two small slits in an opaque screen, illuminated by the same monochromatic light source. We’ll see that if there were not a constant phase relationship between the two sources, the phenomena we are about to discuss would not occur. Two monochromatic sources of the same frequency and with a constant phase relationship (not necessarily in phase) are said to be **coherent**. We also use the term *coherent waves* (or, for light waves, *coherent light*) to refer to the waves emitted by two such sources.

Figure 35.1 A “snapshot” of sinusoidal waves of frequency f and wavelength λ spreading out from source S_1 in all directions.

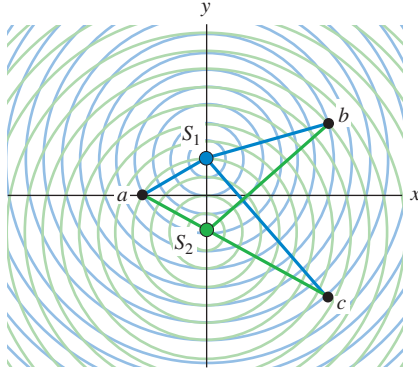
Wave fronts: crests of the wave (frequency f) separated by one wavelength λ



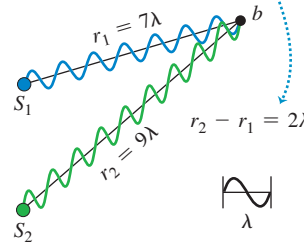
The wave fronts move outward from source S_1 at the wave speed $v = f\lambda$.

Figure 35.2 (a) A “snapshot” of sinusoidal waves spreading out from two coherent sources S_1 and S_2 . Constructive interference occurs at point a (equidistant from the two sources) and (b) at point b . (c) Destructive interference occurs at point c .

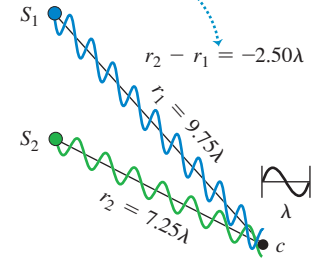
(a) Two coherent wave sources separated by a distance 4λ



(b) Conditions for constructive interference: Waves interfere constructively if their path lengths differ by an integer number of wavelengths: $r_2 - r_1 = m\lambda$.



(c) Conditions for destructive interference: Waves interfere destructively if their path lengths differ by a half-integer number of wavelengths: $r_2 - r_1 = (m + \frac{1}{2})\lambda$.



If the waves emitted by the two coherent sources are *transverse*, like electromagnetic waves, then we'll also assume that the wave disturbances produced by both sources have the same *polarization* (that is, they lie along the same line). For example, the sources S_1 and S_2 in Fig. 35.2a could be two radio antennas in the form of long rods oriented parallel to the z -axis (perpendicular to the plane of the figure); at any point in the xy -plane the waves produced by both antennas have $\vec{E}$ fields with only a z -component. Then we need only a single scalar function to describe each wave; this makes the analysis much easier.

We position the two sources of equal amplitude, equal wavelength, and (if the waves are transverse) the same polarization along the z -axis in Fig. 35.2a, equidistant from the origin. Consider a point a on the x -axis. From symmetry the two distances from S_1 to a and from S_2 to a are *equal*; waves from the two sources thus require equal times to travel to a . Hence waves that leave S_1 and S_2 in phase arrive at a in phase, and the total amplitude at a is *twice* the amplitude of each individual wave. This is true for *any* point on the x -axis.

Similarly, the distance from S_2 to point b is exactly two wavelengths *greater* than the distance from S_1 to b . A wave crest from S_1 arrives at b exactly two cycles earlier than a crest emitted at the same time from S_2 , and again the two waves arrive in phase. As at point a , the total amplitude is the sum of the amplitudes of the waves from S_1 and S_2 .

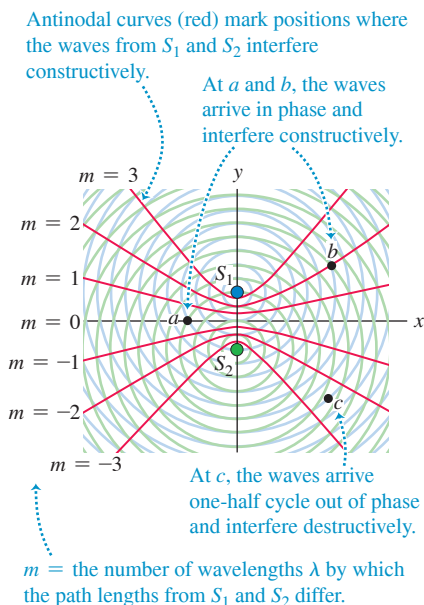
In general, when waves from two or more sources arrive at a point *in phase*, they reinforce each other: The amplitude of the resultant wave is the *sum* of the amplitudes of the individual waves. This is called **constructive interference** (Fig. 35.2b). Let the distance from S_1 to any point P be r_1 , and let the distance from S_2 to P be r_2 . For constructive interference to occur at P , the path difference $r_2 - r_1$ for the two sources must be an integer multiple of the wavelength λ :

$$r_2 - r_1 = m\lambda \quad (m = 0, \pm 1, \pm 2, \pm 3, \dots) \quad \begin{array}{l} \text{(constructive} \\ \text{interference,} \\ \text{sources in phase)} \end{array} \quad (35.1)$$

In Fig. 35.2a, points a and b satisfy Eq. (35.1) with $m = 0$ and $m = +2$, respectively.

Something different occurs at point c in Fig. 35.2a. At this point, the path difference $r_2 - r_1 = -2.50\lambda$, which is a *half-integer* number of wavelengths. Waves from the two sources arrive at point c exactly a half-cycle out of phase. A crest of one wave arrives at the same time as a crest in the opposite direction (a “trough”) of the other wave (Fig. 35.2c). The resultant amplitude is the *difference* between the two individual amplitudes. If the individual amplitudes are equal, then the total amplitude is *zero*! This cancellation or partial

Figure 35.3 The same as Fig. 35.2a, but with red antinodal curves (curves of maximum amplitude) superimposed. All points on each curve satisfy Eq. (35.1) with the value of m shown. The nodal curves (not shown) lie between each adjacent pair of antinodal curves.



BIO APPLICATION **Phase Difference, Path Difference, and Localization in Human Hearing** Your auditory system uses the phase differences between sounds received by your left and your right ears for *localization*—determining the direction from which the sounds are coming. For sound waves with frequencies lower than about 800 Hz (which are important in speech and music), the distance between your ears is less than a half-wavelength and the phase difference between the waves detected by each ear is less than a half cycle. Remarkably, your brain can detect this phase difference, determine the corresponding path difference, and use this information to localize the direction of the sound source.



cancellation of the individual waves is called **destructive interference**. The condition for destructive interference in the situation shown in Fig. 35.2a is

$$r_2 - r_1 = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \pm 3, \dots) \quad \begin{array}{l} \text{(destructive} \\ \text{interference,} \\ \text{sources in phase)} \end{array} \quad (35.2)$$

The path difference at point c in Fig. 35.2a satisfies Eq. (35.2) with $m = -3$.

Figure 35.3 shows the same situation as in Fig. 35.2a, but with red curves that show all positions where *constructive* interference occurs. On each curve, the path difference $r_2 - r_1$ is equal to an integer m times the wavelength, as in Eq. (35.1). These curves are called **antinodal curves**. They are directly analogous to *antinodes* in the standing-wave patterns described in Chapters 15 and 16 and Section 32.5. In a standing wave formed by interference between waves propagating in opposite directions, the antinodes are points at which the amplitude is maximum; likewise, the wave amplitude in the situation of Fig. 35.3 is maximum along the antinodal curves. Not shown in Fig. 35.3 are the **nodal curves**, which are the curves that show where *destructive* interference occurs in accordance with Eq. (35.2); these are analogous to the *nodes* in a standing-wave pattern. A nodal curve lies between each two adjacent antinodal curves in Fig. 35.3; one such curve, corresponding to $r_2 - r_1 = -2.50\lambda$, passes through point c .

In some cases, such as two loudspeakers or two radio-transmitter antennas, the interference pattern is three-dimensional. Think of rotating the color curves of Fig. 35.3 around the y -axis; then maximum constructive interference occurs at all points on the resulting surfaces of revolution.

CAUTION **Interference patterns are not standing waves** In the standing waves described in Sections 15.7, 16.4, and 32.5, the interference is between two waves propagating in opposite directions; there is *no* net energy flow in either direction (the energy in the wave is left “standing”). In the situations shown in Figs. 35.2a and 35.3, there is likewise a stationary pattern of antinodal and nodal curves, but there is a net flow of energy *outward* from the two sources. All that interference does is to “channel” the energy flow so that it is greatest along the antinodal curves and least along the nodal curves. ■

For Eqs. (35.1) and (35.2) to hold, the two sources must be monochromatic, must have the same wavelength, and must *always* be in phase. These conditions are rather easy to satisfy for sound waves. But with *light* waves there is no practical way to achieve a constant phase relationship (coherence) with two independent sources. This is because of the way light is emitted. In ordinary light sources, atoms gain excess energy by thermal agitation or by impact with accelerated electrons. Such an “excited” atom begins to radiate energy and continues until it has lost all the energy it can, typically in a time of the order of 10^{-8} s. The many atoms in a source ordinarily radiate in an unsynchronized and random phase relationship, and the light that is emitted from *two* such sources has no definite phase relationship.

However, the light from a single source can be split so that parts of it emerge from two or more regions of space, forming two or more *secondary sources*. Then any random phase change in the source affects these secondary sources equally and does not change their *relative* phase.

The distinguishing feature of light from a *laser* is that the emission of light from many atoms is synchronized in frequency and phase. As a result, the random phase changes mentioned above occur much less frequently. Definite phase relationships are preserved over correspondingly much greater lengths in the beam, and laser light is much more coherent than ordinary light.

TEST YOUR UNDERSTANDING OF SECTION 35.1 Consider a point in Fig. 35.3 on the positive y -axis above S_1 . Does this point lie on (i) an antinodal curve; (ii) a nodal curve; or (iii) neither? (Hint: The distance between S_1 and S_2 is 4λ .)

ANSWER ■ (i) At any point P on the positive y -axis above S_1 , the distance r_2 from S_2 to P is greater than the distance r_1 from S_1 to P by 4λ . This corresponds to $m = 4$ in Eq. (35.1), the equation for constructive interference. Hence all such points make up an antinodal curve.

35.2 TWO-SOURCE INTERFERENCE OF LIGHT

The interference pattern produced by two coherent sources of *water* waves of the same wavelength can be readily seen in a ripple tank with a shallow layer of water (**Fig. 35.4**). This pattern is not directly visible when the interference is between *light* waves, since light traveling in a uniform medium cannot be seen. (Sunlight in a room is made visible by scattering from airborne dust.)

Figure 35.5a shows one of the earliest quantitative experiments to reveal the interference of light from two sources, first performed in 1800 by the English scientist Thomas Young. Let's examine this important experiment in detail. A light source (not shown) emits monochromatic light; however, this light is not suitable for use in an interference experiment because emissions from different parts of an ordinary source are not synchronized. To remedy this, the light is directed at a screen with a narrow slit S_0 , $1\text{ }\mu\text{m}$ or so wide. The light emerging from the slit originated from only a small region of the light source; thus slit S_0 behaves more nearly like the idealized source shown in Fig. 35.1. (In modern versions of the experiment, a laser is used as a source of coherent light, and the slit S_0 isn't needed.) The light from slit S_0 falls on a screen with two other narrow slits S_1 and S_2 , each $1\text{ }\mu\text{m}$ or so wide and a few tens or hundreds of micrometers apart. Cylindrical wave fronts spread out from slit S_0 and reach slits S_1 and S_2 *in phase* because they travel equal distances from S_0 . The waves *emerging* from slits S_1 and S_2 are therefore also always in phase, so S_1 and S_2 are *coherent* sources. The interference of waves from S_1 and S_2 produces a pattern in space like that to the right of the sources in Figs. 35.2a and 35.3.

To visualize the interference pattern, a screen is placed so that the light from S_1 and S_2 falls on it (Fig. 35.5b). The screen will be most brightly illuminated at points P , where the light waves from the slits interfere constructively, and will be darkest at points where the interference is destructive.

Figure 35.4 The concepts of constructive interference and destructive interference apply to these water waves as well as to light waves and sound waves.

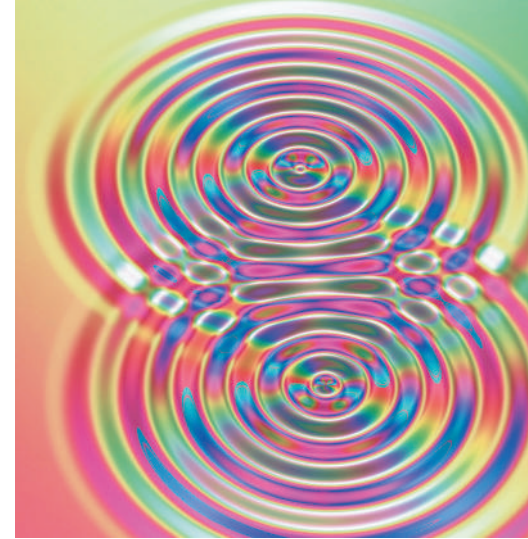
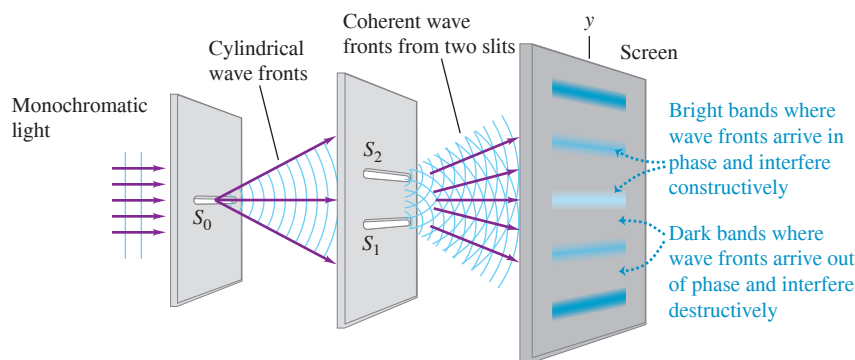
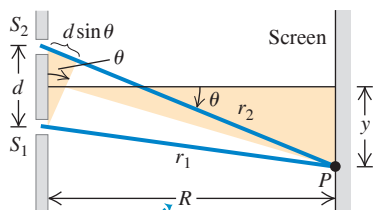


Figure 35.5 (a) Young's experiment to show interference of light passing through two slits. A pattern of bright and dark areas appears on the screen (see Fig. 35.6). (b) Geometrical analysis of Young's experiment. For the case shown, $r_2 > r_1$ and both y and θ are positive. If point P is on the other side of the screen's center, $r_2 < r_1$ and both y and θ are negative. (c) Approximate geometry when the distance R to the screen is much greater than the distance d between the slits.

(a) Interference of light waves passing through two slits

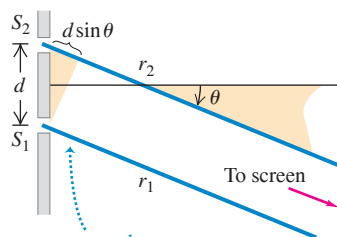


(b) Actual geometry (seen from the side)



In real situations, the distance R to the screen is usually very much greater than the distance d between the slits ...

(c) Approximate geometry



... so we can treat the rays as parallel, in which case the path difference is simply $r_2 - r_1 = d \sin \theta$.

To simplify the analysis of Young's experiment, we assume that the distance R from the slits to the screen is so large in comparison to the distance d between the slits that the lines from S_1 and S_2 to P are very nearly parallel, as in Fig. 35.5c. This is usually the case for experiments with light; the slit separation is typically a few millimeters, while the screen may be a meter or more away. The difference in path length is then given by

$$r_2 - r_1 = d \sin \theta \quad (35.3)$$

where θ is the angle between a line from slits to screen (shown in blue in Fig. 35.5c) and the normal to the plane of the slits (shown as a thin black line).

Constructive and Destructive Two-Slit Interference

We found in Section 35.1 that constructive interference (reinforcement) occurs at points where the path difference is an integer number of wavelengths, $m\lambda$, where $m = 0, \pm 1, \pm 2, \pm 3, \dots$. So the bright regions on the screen in Fig. 35.5a occur at angles θ for which

Constructive interference, two slits:

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.4)$$

Distance between slits Wavelength
Angle of line from slits to m th bright region on screen

Similarly, destructive interference (cancellation) occurs, forming dark regions on the screen, at points for which the path difference is a half-integer number of wavelengths, $(m + \frac{1}{2})\lambda$:

Destructive interference, two slits:

$$d \sin \theta = (m + \frac{1}{2})\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.5)$$

Distance between slits Wavelength
Angle of line from slits to m th dark region on screen

Thus the pattern on the screen of Figs. 35.5a and 35.5b is a succession of bright and dark bands, or **interference fringes**, parallel to the slits S_1 and S_2 . A photograph of such a pattern is shown in Fig. 35.6.

We can derive an expression for the positions of the centers of the bright bands on the screen. In Fig. 35.5b, y is measured from the center of the pattern, corresponding to the distance from the center of Fig. 35.6. Let y_m be the distance from the center of the pattern ($\theta = 0$) to the center of the m th bright band. Let θ_m be the corresponding value of θ ; then

$$y_m = R \tan \theta_m$$

In such experiments, the distances y_m are often much smaller than the distance R from the slits to the screen. Hence θ_m is very small, $\tan \theta_m \approx \sin \theta_m$, and

$$y_m = R \sin \theta_m$$

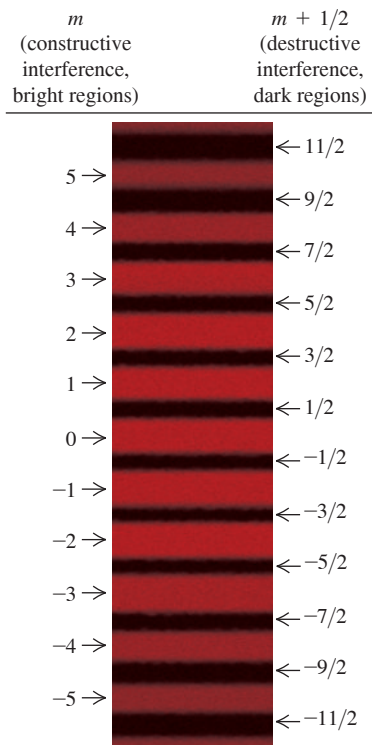
Combining this with Eq. (35.4), we find that for *small angles only*,

Constructive interference, Young's experiment (small angles only):

$$y_m = R \frac{m\lambda}{d} \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.6)$$

Position of m th bright band Wavelength
Distance from slits to screen Distance between slits

Figure 35.6 Photograph of interference fringes produced on a screen in Young's double-slit experiment. The center of the pattern is a bright band corresponding to $m = 0$ in Eq. (35.4); this point on the screen is equidistant from the two slits.



CAUTION Equation (35.6) is for **small angles only**. While Eqs. (35.4) and (35.5) are valid for any angle, Eq. (35.6) is valid for *small* angles only. It can be used *only* if the distance R from slits to screen is much greater than the slit separation d and if R is much greater than the distance y_m from the center of the interference pattern to the m th bright band. ■

We can measure R and d , as well as the positions y_m of the bright fringes, so this experiment provides a direct measurement of the wavelength λ . Young's experiment was in fact the first direct measurement of wavelengths of light.

The distance between adjacent bright bands in the pattern is *inversely* proportional to the distance d between the slits. The closer together the slits are, the more the pattern spreads out. When the slits are far apart, the bands in the pattern are closer together.

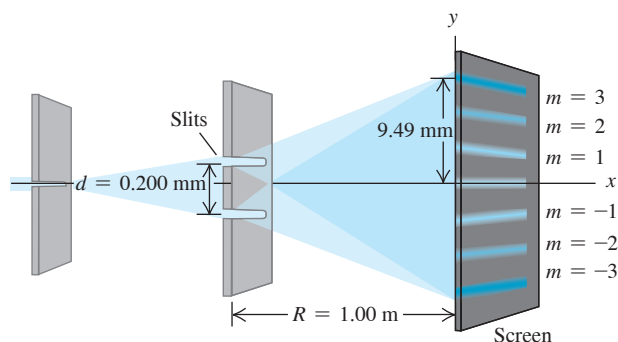
While we have described the experiment that Young performed with visible light, the results given in Eqs. (35.4) and (35.5) are valid for *any* type of wave, provided that the resultant wave from two coherent sources is detected at a point that is far away in comparison to the separation d .

EXAMPLE 35.1 Two-slit interference

WITH  **ARIATION PROBLEMS**

Figure 35.7 shows a two-slit interference experiment in which the slits are 0.200 mm apart and the screen is 1.00 m from the slits. The $m = 3$ bright fringe in the figure is 9.49 mm from the central bright fringe. Find the wavelength of the light.

Figure 35.7 Using a two-slit interference experiment to measure the wavelength of light.



IDENTIFY and SET UP Our target variable in this two-slit interference problem is the wavelength λ . We are given the slit separation $d = 0.200$ mm, the distance from slits to screen $R = 1.00$ m, and the distance $y_3 = 9.49$ mm on the screen from the center of the interference pattern to the $m = 3$ bright fringe. We may use Eq. (35.6) to find λ , since the value of R is so much greater than the value of d or y_3 .

EXECUTE We solve Eq. (35.6) for λ for the case $m = 3$:

$$\lambda = \frac{y_m d}{m R} = \frac{(9.49 \times 10^{-3} \text{ m})(0.200 \times 10^{-3} \text{ m})}{(3)(1.00 \text{ m})} = 633 \times 10^{-9} \text{ m} = 633 \text{ nm}$$

EVALUATE This bright fringe could also correspond to $m = -3$. Can you show that this gives the same result for λ ?

KEYCONCEPT In two-slit interference, each wave front of a monochromatic wave enters the two slits simultaneously. Hence the waves that emerge from the slits always have the same phase relationship and so produce a steady interference pattern. Equation (35.6) gives the positions of the bright fringes where constructive interference occurs.

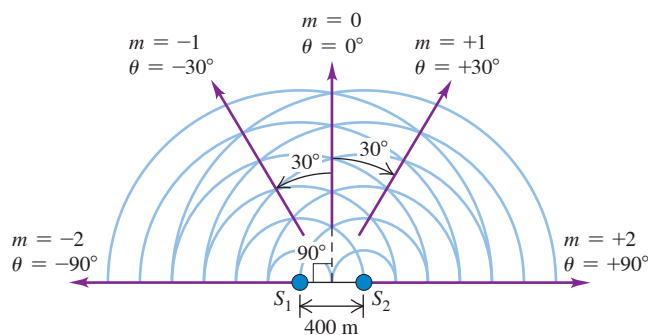
EXAMPLE 35.2 Broadcast pattern of a radio station

WITH  **ARIATION PROBLEMS**

It is often desirable to radiate most of the energy from a radio transmitter in particular directions rather than uniformly in all directions. Pairs or rows of antennas are often used to produce the desired radiation pattern. As an example, consider two identical vertical antennas 400 m apart, operating at 1500 kHz = 1.5×10^6 Hz (near the top end of the AM broadcast band) and oscillating in phase. At distances much greater than 400 m, in what directions is the intensity from the two antennas greatest?

IDENTIFY and SET UP The antennas, shown in **Fig. 35.8**, correspond to sources S_1 and S_2 in Fig. 35.5. Hence we can apply the ideas of two-slit interference to this problem. Since the resultant wave is detected at distances much greater than $d = 400$ m, we may use Eq. (35.4) to give the directions of the intensity maxima, the values of θ for which the path difference is zero or a whole number of wavelengths.

Figure 35.8 Two radio antennas broadcasting in phase. The purple arrows indicate the directions of maximum intensity. The waves that are emitted toward the lower half of the figure are not shown.



EXECUTE The wavelength is $\lambda = c/f = 200$ m. From Eq. (35.4) with $m = 0, \pm 1$, and ± 2 , the intensity maxima are given by

$$\sin \theta = \frac{m\lambda}{d} = \frac{m(200 \text{ m})}{400 \text{ m}} = \frac{m}{2} \quad \theta = 0, \pm 30^\circ, \pm 90^\circ$$

In this example, values of m greater than 2 or less than -2 give values of $\sin \theta$ greater than 1 or less than -1 , which is impossible. There is *no* direction for which the path difference is three or more wavelengths, so values of m of ± 3 or beyond have no meaning in this example.

EVALUATE We can check our result by calculating the angles for *minimum* intensity, using Eq. (35.5). There should be one intensity minimum between each pair of intensity maxima, just as in Fig. 35.6. From Eq. (35.5), with $m = -2, -1, 0$, and 1 ,

$$\sin \theta = \frac{(m + \frac{1}{2})\lambda}{d} = \frac{m + \frac{1}{2}}{2} \quad \theta = \pm 14.5^\circ, \pm 48.6^\circ$$

These angles fall between the angles for intensity maxima, as they should. The angles are not small, so the angles for the minima are *not* exactly halfway between the angles for the maxima.

KEYCONCEPT Equations (35.4) and (35.5) give the angles in two-slit interference for which constructive interference (bright regions) and destructive interference (dark regions) occur. These equations are also valid for any two monochromatic sources that emit in phase.

TEST YOUR UNDERSTANDING OF SECTION 35.2 You shine a tunable laser (whose wavelength can be adjusted by turning a knob) on a pair of closely spaced slits. The light emerging from the two slits produces an interference pattern on a screen like that shown in Fig. 35.6. If you adjust the wavelength so that the laser light changes from red to blue, how will the spacing between bright fringes change? (i) The spacing increases; (ii) the spacing decreases; (iii) the spacing is unchanged; (iv) not enough information is given to decide.

ANSWER

(ii) Blue light has a shorter wavelength than red light (see Section 32.1). Equation (35.6) tells us that the distance y_m from the center of the pattern to the m th bright fringe is proportional to the wavelength λ . Hence all of the fringes will move toward the center of the pattern as the wavelength decreases, and the spacing between fringes will decrease.

35.3 INTENSITY IN INTERFERENCE PATTERNS

In Section 35.2 we found the positions of maximum and minimum intensity in a two-source interference pattern. Let's now see how to find the intensity at *any* point in the pattern. To do this, we have to combine the two sinusoidally varying fields (from the two sources) at a point P in the radiation pattern, taking proper account of the phase difference of the two waves at point P , which results from the path difference. The intensity is then proportional to the square of the resultant electric-field amplitude, as we learned in Section 32.4.

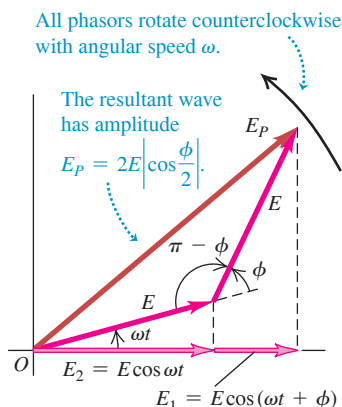
To calculate the intensity, we'll assume, as in Section 35.2, that the waves from the two sources have equal amplitude E and the same polarization. This assumes that the sources are identical and ignores the slight amplitude difference caused by the unequal path lengths (the amplitude decreases with increasing distance from the source). From Eq. (32.29), each source by itself would give an intensity $\frac{1}{2}\epsilon_0 c E^2$ at point P . If the two sources are in phase, then the waves that arrive at P differ in phase by an amount ϕ that is proportional to the difference in their path lengths, $(r_2 - r_1)$. Then we can use the following expressions for the two electric fields superposed at P :

$$E_1(t) = E \cos(\omega t + \phi)$$

$$E_2(t) = E \cos \omega t$$

The superposition of the two fields at P is a sinusoidal function with some amplitude E_P that depends on E and the phase difference ϕ . First we'll work on finding the amplitude E_P if E and ϕ are known. Then we'll find the intensity I of the resultant wave, which is proportional to E_P^2 . Finally, we'll relate the phase difference ϕ to the path difference, which is determined by the geometry of the situation.

Figure 35.9 Phasor diagram for the superposition at a point P of two waves of equal amplitude E with a phase difference ϕ .



Amplitude in Two-Source Interference

To add the two sinusoidal functions with a phase difference, we use the same *phasor* representation that we used for simple harmonic motion (see Section 14.2) and for voltages and currents in ac circuits (see Section 31.1). We suggest that you review these sections now. Each sinusoidal function is represented by a rotating vector (phasor) whose projection on the horizontal axis at any instant represents the instantaneous value of the sinusoidal function.

In Fig. 35.9, E_1 is the horizontal component of the phasor representing the wave from source S_1 , and E_2 is the horizontal component of the phasor for the wave from S_2 . As shown in the diagram, both phasors have the same magnitude E , but E_1 is *ahead* of E_2 in phase by an angle ϕ . Both phasors rotate counterclockwise with constant angular speed ω , and the sum of the projections on the horizontal axis at any time gives the instantaneous value of the total E field at point P . Thus the amplitude E_P of the resultant sinusoidal wave at P is the magnitude of the dark red phasor in the diagram (labeled E_P); this is the *vector sum*

of the other two phasors. To find E_P , we use the law of cosines and the trigonometric identity $\cos(\pi - \phi) = -\cos \phi$:

$$\begin{aligned} E_P^2 &= E^2 + E^2 - 2E^2 \cos(\pi - \phi) \\ &= E^2 + E^2 + 2E^2 \cos \phi \end{aligned}$$

Then, using the identity $1 + \cos \phi = 2\cos^2(\phi/2)$, we obtain

$$E_P^2 = 2E^2(1 + \cos \phi) = 4E^2 \cos^2\left(\frac{\phi}{2}\right)$$

$$E_P = 2E \left| \cos \frac{\phi}{2} \right| \quad (35.7)$$

You can also obtain this result without using phasors.

When the two waves are in phase, $\phi = 0$ and $E_P = 2E$. When they are exactly a half-cycle out of phase, $\phi = \pi \text{ rad} = 180^\circ$, $\cos(\phi/2) = \cos(\pi/2) = 0$, and $E_P = 0$. Thus the superposition of two sinusoidal waves with the same frequency and amplitude but with a phase difference yields a sinusoidal wave with the same frequency and an amplitude between zero and twice the individual amplitudes, depending on the phase difference.

Intensity in Two-Source Interference

To obtain the intensity I at point P , we recall from Section 32.4 that I is equal to the average magnitude of the Poynting vector, S_{av} . For a sinusoidal wave with electric-field amplitude E_P , this is given by Eq. (32.29) with E_{max} replaced by E_P . Thus we can express the intensity in several equivalent forms:

$$I = S_{\text{av}} = \frac{E_P^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_P^2 = \frac{1}{2} \epsilon_0 c E_P^2 \quad (35.8)$$

The essential content of these expressions is that I is proportional to E_P^2 . When we substitute Eq. (35.7) into the last expression in Eq. (35.8), we get

$$I = \frac{1}{2} \epsilon_0 c E_P^2 = 2\epsilon_0 c E^2 \cos^2 \frac{\phi}{2} \quad (35.9)$$

In particular, the *maximum* intensity I_0 , which occurs at points where the phase difference is zero ($\phi = 0$), is

$$I_0 = 2\epsilon_0 c E^2$$

Note that the maximum intensity I_0 is *four times* (not twice) as great as the intensity $\frac{1}{2} \epsilon_0 c E^2$ from each individual source. Substituting the expression for I_0 into Eq. (35.9), we find

$$I = I_0 \cos^2 \frac{\phi}{2} \quad (35.10)$$

The intensity depends on the phase difference ϕ and varies between I_0 and zero. If we average Eq. (35.10) over all possible phase differences, the result is $I_0/2 = \epsilon_0 c E^2$ [the average of $\cos^2(\phi/2)$ is $\frac{1}{2}$]. This is just twice the intensity from each individual source, as we should expect. The total energy output from the two sources isn't changed by the interference effects, but the energy is redistributed (see Section 35.1).

Phase Difference and Path Difference

Our next task is to find the phase difference ϕ between the two fields at any point P . We know that ϕ is proportional to the difference in path length from the two sources to point P . When the path difference is one wavelength, the phase difference is one cycle, and $\phi = 2\pi \text{ rad} = 360^\circ$. When the path difference is $\lambda/2$, $\phi = \pi \text{ rad} = 180^\circ$, and so on. That is, the ratio of the phase difference ϕ to 2π is equal to the ratio of the path difference $r_2 - r_1$ to λ :

$$\frac{\phi}{2\pi} = \frac{r_2 - r_1}{\lambda}$$

$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1) \quad (35.11)$$

We introduced the wave number $k = 2\pi/\lambda$ in Section 15.3.

If the material in the space between the sources and P is anything other than vacuum, we must use the wavelength *in the material* in Eq. (35.11). If λ_0 and k_0 are the wavelength and wave number, respectively, in vacuum and the material has index of refraction n , then

$$\lambda = \frac{\lambda_0}{n} \quad \text{and} \quad k = nk_0 \quad (35.12)$$

Finally, if the point P is far away from the sources in comparison to their separation d , the path difference is given by Eq. (35.3):

$$r_2 - r_1 = d \sin \theta$$

Combining this with Eq. (35.11), we find

$$\phi = k(r_2 - r_1) = kd \sin \theta = \frac{2\pi d}{\lambda} \sin \theta \quad (35.13)$$

When we substitute this into Eq. (35.10), we find

$$I = I_0 \cos^2\left(\frac{1}{2}kd \sin \theta\right) = I_0 \cos^2\left(\frac{\pi d}{\lambda} \sin \theta\right) \quad \text{(intensity far from two sources)} \quad (35.14)$$

Maximum intensity occurs when the cosine has the values ± 1 —that is, when

$$\frac{\pi d}{\lambda} \sin \theta = m\pi \quad (m = 0, \pm 1, \pm 2, \dots)$$

or

$$d \sin \theta = m\lambda$$

in agreement with Eq. (35.4). You can also derive Eq. (35.5) for the zero-intensity directions from Eq. (35.14).

As we noted in Section 35.2, in experiments with light we visualize the interference pattern due to two slits by using a screen placed at a distance R from the slits. We can describe positions on the screen with the coordinate y ; the positions of the bright fringes are given by Eq. (35.6), where ordinarily $y \ll R$. In that case, $\sin \theta$ is approximately equal to y/R , and we obtain the following expressions for the intensity at *any* point on the screen as a function of y :

$$I = I_0 \cos^2\left(\frac{kdy}{2R}\right) = I_0 \cos^2\left(\frac{\pi dy}{\lambda R}\right) \quad \text{(intensity in two-slit interference)} \quad (35.15)$$

Figure 35.10 Intensity distribution in the interference pattern from two identical slits.

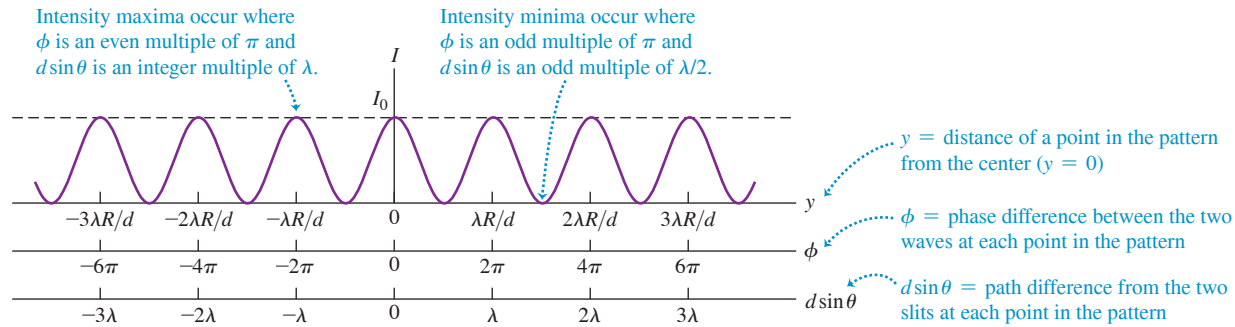


Figure 35.10 shows a graph of Eq. (35.15); we can compare this with the photographically recorded pattern of Fig. 35.6. All peaks in Fig. 35.10 have the same intensity, while those in Fig. 35.6 fade off as we go away from the center. We'll explore the reasons for this variation in peak intensity in Chapter 36.

EXAMPLE 35.3 A directional transmitting antenna array

WITH VARIATION PROBLEMS

Suppose the two identical radio antennas of Fig. 35.8 are moved to be only 10.0 m apart and the broadcast frequency is increased to $f = 60.0$ MHz. At a distance of 700 m from the point midway between the antennas and in the direction $\theta = 0$ (see Fig. 35.8), the intensity is $I_0 = 0.020$ W/m². At this same distance, find (a) the intensity in the direction $\theta = 4.0^\circ$; (b) the direction near $\theta = 0$ for which the intensity is $I_0/2$; and (c) the directions in which the intensity is zero.

IDENTIFY and SET UP This problem involves the intensity distribution as a function of angle. Because the 700 m distance from the antennas to the point at which the intensity is measured is much greater than the distance $d = 10.0$ m between the antennas, the amplitudes of the waves from the two antennas are very nearly equal. Hence we can use Eq. (35.14) to relate intensity I and angle θ .

EXECUTE The wavelength is $\lambda = c/f = 5.00$ m. The spacing $d = 10.0$ m between the antennas is just twice the wavelength (as was the case in Example 35.2), so $d/\lambda = 2.00$ and Eq. (35.14) becomes

$$I = I_0 \cos^2 \left(\frac{\pi d}{\lambda} \sin \theta \right) = I_0 \cos^2 [(2.00\pi \text{ rad}) \sin \theta]$$

(a) When $\theta = 4.0^\circ$,

$$\begin{aligned} I &= I_0 \cos^2 [(2.00\pi \text{ rad}) \sin 4.0^\circ] = 0.82 I_0 \\ &= (0.82)(0.020 \text{ W/m}^2) = 0.016 \text{ W/m}^2 \end{aligned}$$

(b) The intensity I equals $I_0/2$ when the cosine in Eq. (35.14) has the value $\pm 1/\sqrt{2}$. The smallest angles at which this occurs correspond to $2.00\pi \sin \theta = \pm \pi/4$ rad, so $\sin \theta = \pm (1/8.00) = \pm 0.125$ and $\theta = \pm 7.2^\circ$.

(c) The intensity is zero when $\cos[(2.00\pi \text{ rad}) \sin \theta] = 0$. This occurs for $2.00\pi \sin \theta = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \dots$, or $\sin \theta = \pm 0.250, \pm 0.750, \pm 1.25, \dots$. Values of $\sin \theta$ greater than 1 have no meaning, so the answers are

$$\theta = \pm 14.5^\circ, \pm 48.6^\circ$$

EVALUATE The condition in part (b) that $I = I_0/2$, so that $(2.00\pi \text{ rad}) \sin \theta = \pm \pi/4$ rad, is also satisfied when $\sin \theta = \pm 0.375, \pm 0.625$, or ± 0.875 so that $\theta = \pm 22.0^\circ, \pm 38.7^\circ$, or $\pm 61.0^\circ$. (Can you verify this?) It would be incorrect to include these angles in the solution, however, because the problem asked for the angle near $\theta = 0$ at which $I = I_0/2$. These additional values of θ aren't the ones we're looking for.

KEYCONCEPT The intensity at a given point in a two-source interference pattern depends on the phase difference ϕ between the waves that arrive at that point from the two sources [Eqs. (35.10) and (35.11)]. The intensity is *maximum* at points where ϕ is an *even* multiple of π ($0, \pm 2\pi, \pm 4\pi, \dots$) and *minimum* at point where ϕ is an *odd* multiple of π ($\pm \pi, \pm 3\pi, \dots$).

TEST YOUR UNDERSTANDING OF SECTION 35.3 A two-slit interference experiment uses coherent light of wavelength 5.00×10^{-7} m. Rank the following points in the interference pattern according to the intensity at each point, from highest to lowest. (i) A point that is closer to one slit than the other by 4.00×10^{-7} m; (ii) a point where the light waves received from the two slits are out of phase by 4.00 rad; (iii) a point that is closer to one slit than the other by 7.50×10^{-7} m; (iv) a point where the light waves received by the two slits are out of phase by 2.00 rad.

ANSWER

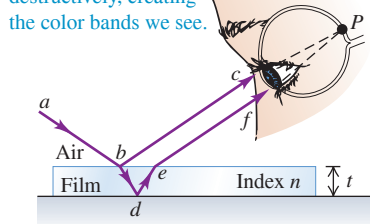
(i) $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (4.00 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(\pi/2) = 0$.
(ii) $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (7.50 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(3\pi/2) = 0$.
(iii) $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (4.00 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(\pi/2) = 0$.
(iv) $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (7.50 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(3\pi/2) = 0$.
Hence we use Eq. (35.14), $I = I_0 \cos^2(\phi/2)$, and we find ϕ and use Eq. (35.10), $\phi = 2\pi d \sin \theta / \lambda$. In parts (ii) and (iii) we are given the phase difference ϕ and use Eq. (35.10), $\phi = 2\pi d \sin \theta / \lambda$. In cases (i) and (iv) we are given the wavelength λ and path difference $d \sin \theta$.
Ranking: (i) and (iv) are highest, followed by (ii) and (iii), which are lowest.

Figure 35.11 (a) A diagram and (b) a photograph showing interference of light reflected from a thin film.

(a) Interference between rays reflected from the two surfaces of a thin film

Light reflected from the upper and lower surfaces of the film comes together in the eye at P and undergoes interference.

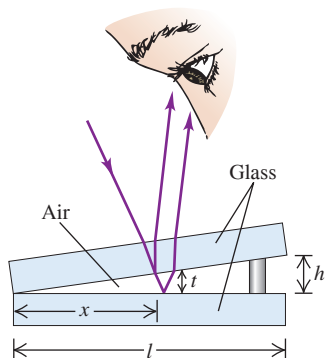
Some colors interfere constructively and others destructively, creating the color bands we see.



(b) Colorful reflections from a soap bubble



Figure 35.12 Interference between light waves reflected from the two sides of an air wedge separating two glass plates. The angles and the thickness of the air wedge have been exaggerated for clarity; in the text we assume that the light strikes the upper plate at normal incidence and that the distances h and t are much less than l .



35.4 INTERFERENCE IN THIN FILMS

You often see bright bands of color when light reflects from a thin layer of oil floating on water or from a soap bubble (see the photograph that opens this chapter). These are the results of interference. Light waves are reflected from the front and back surfaces of such thin films, and constructive interference between the two reflected waves (with different path lengths) occurs in different places for different wavelengths. **Figure 35.11a** shows the situation. Light shining on the upper surface of a thin film with thickness t is partly reflected at the upper surface (path abc). Light *transmitted* through the upper surface is partly reflected at the lower surface (path $abdef$). The two reflected waves come together at point P on the retina of the eye. Depending on the phase relationship, they may interfere constructively or destructively. Different colors have different wavelengths, so the interference may be constructive for some colors and destructive for others. That's why we see colored patterns in the photograph that opens this chapter (which shows a thin film of oil floating on water) and in Fig. 35.11b (which shows thin films of soap solution that make up the bubble walls). The complex shapes of the colored patterns result from variations in the thickness of the film.

Thin-Film Interference and Phase Shifts During Reflection

Let's look at a simplified situation in which *monochromatic* light reflects from two nearly parallel surfaces at nearly normal incidence. **Figure 35.12** shows two plates of glass separated by a thin wedge, or film, of air. We want to consider interference between the two light waves reflected from the surfaces adjacent to the air wedge. (Reflections also occur at the top surface of the upper plate and the bottom surface of the lower plate; to keep our discussion simple, we won't include these.) The situation is the same as in Fig. 35.11a except that the film (wedge) thickness is not uniform. The path difference between the two waves is just twice the thickness t of the air wedge at each point. At points where $2t$ is an integer number of wavelengths, we expect to see constructive interference and a bright area; where it is a half-integer number of wavelengths, we expect to see destructive interference and a dark area. Where the plates are in contact, there is practically *no* path difference, and we expect a bright area.

When we carry out the experiment, the bright and dark fringes appear, but they are interchanged! Along the line where the plates are in contact, we find a *dark* fringe, not a bright one. This suggests that one or the other of the reflected waves has undergone a half-cycle phase shift during its reflection. In that case the two waves that are reflected at the line of contact are a half-cycle out of phase even though they have the same path length.

In fact, this phase shift can be predicted from Maxwell's equations and the electromagnetic nature of light. The details of the derivation are beyond our scope, but here is the result. Suppose a light wave with electric-field amplitude E_i is traveling in an optical material with index of refraction n_a . It strikes, at normal incidence, an interface with another optical material with index n_b . The amplitude E_r of the wave reflected from the interface is given by

$$E_r = \frac{n_a - n_b}{n_a + n_b} E_i \quad (\text{normal incidence}) \quad (35.16)$$

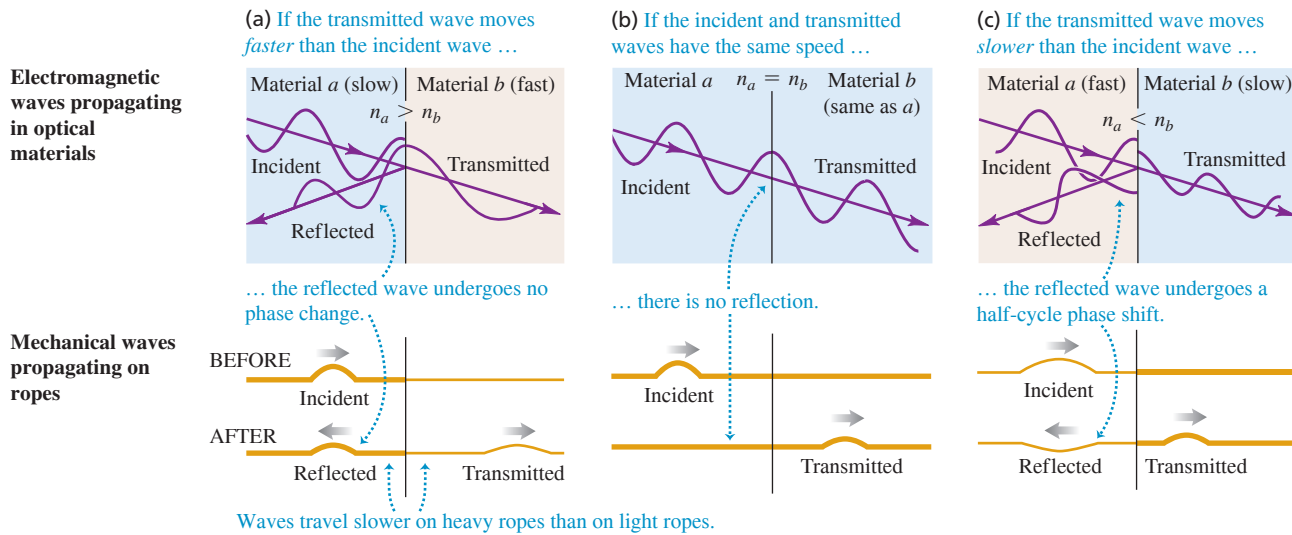
This result shows that the incident and reflected amplitudes have the same sign when n_a is larger than n_b and opposite signs when n_b is larger than n_a . Because amplitudes must always be positive or zero, a *negative* value means that the wave actually undergoes a half-cycle (180°) phase shift. **Figure 35.13** shows three possibilities:

Figure 35.13a: When $n_a > n_b$, light travels more slowly in the first material than in the second. In this case, E_r and E_i have the same sign, and the phase shift of the reflected wave relative to the incident wave is zero. This is analogous to reflection of a transverse mechanical wave on a heavy rope at a point where it is tied to a lighter rope.

Figure 35.13b: When $n_a = n_b$, the amplitude E_r of the reflected wave is zero. In effect there is no interface, so there is *no* reflected wave.

Figure 35.13c: When $n_a < n_b$, light travels more slowly in the second material than in the first. In this case, E_r and E_i have opposite signs, and the phase shift of the reflected wave relative to the incident wave is π rad (a half-cycle). This is analogous to reflection (with inversion) of a transverse mechanical wave on a light rope at a point where it is tied to a heavier rope.

Figure 35.13 Upper figures: electromagnetic waves striking an interface between optical materials at normal incidence (shown as a small angle for clarity). Lower figures: mechanical wave pulses on ropes.



Let's compare with the situation of Fig. 35.12. For the wave reflected from the upper surface of the air wedge, n_a (glass) is greater than n_b , so this wave has zero phase shift. For the wave reflected from the lower surface, n_a (air) is less than n_b (glass), so this wave has a half-cycle phase shift. Waves that are reflected from the line of contact have no path difference to give additional phase shifts, and they interfere destructively; this is what we observe. You can use the above principle to show that for normal incidence, the wave reflected at point b in Fig. 35.11a is shifted by a half-cycle, while the wave reflected at d is not (if there is air below the film).

We can summarize this discussion mathematically. If the film has thickness t , the light is at normal incidence and has wavelength λ in the film; if neither or both of the reflected waves from the two surfaces have a half-cycle reflection phase shift, the conditions for constructive and destructive interference are

Constructive reflection $2t = m\lambda \quad (m = 0, 1, 2, \dots)$ (35.17a)

(From thin film,
no relative phase shift)

Thickness of film $\swarrow$ $\searrow$ Wavelength

Destructive reflection $2t = (m + \frac{1}{2})\lambda \quad (m = 0, 1, 2, \dots)$ (35.17b)

If *one* of the two waves has a half-cycle reflection phase shift, the conditions for constructive and destructive interference are reversed:

Constructive reflection $2t = (m + \frac{1}{2})\lambda \quad (m = 0, 1, 2, \dots)$ (35.18a)

(From thin film,
half-cycle phase shift)

Thickness of film $\swarrow$ $\searrow$ Wavelength

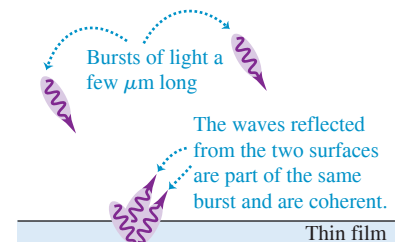
Destructive reflection $2t = m\lambda \quad (m = 0, 1, 2, \dots)$ (35.18b)

Thin and Thick Films

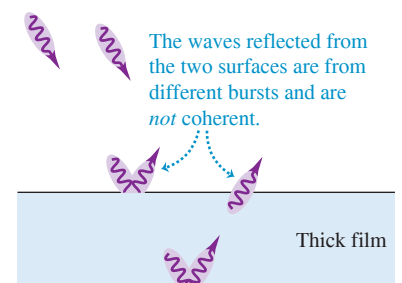
We have emphasized *thin* films in our discussion because of a principle we introduced in Section 35.1: In order for two waves to cause a steady interference pattern, the waves must be *coherent*, with a definite and constant phase relationship. The sun and light bulbs emit light in a stream of short bursts, each of which is only a few micrometers long (1 micrometer = $1 \mu\text{m} = 10^{-6} \text{ m}$). If light reflects from the two surfaces of a thin film, the two reflected waves are part of the same burst (Fig. 35.14a). Hence these waves are coherent and interference occurs as we have described. If the film is too thick, however,

Figure 35.14 (a) Light reflecting from a thin film produces a steady interference pattern, but (b) light reflecting from a thick film does not.

(a) Light reflecting from a thin film



(b) Light reflecting from a thick film



the two reflected waves will belong to different bursts (Fig. 35.14b). There is no definite phase relationship between different light bursts, so the two waves are incoherent and there is no fixed interference pattern. That's why you see interference colors in light reflected from a soap bubble a few micrometers thick (see Fig. 35.11b), but you do *not* see such colors in the light reflected from a pane of window glass with a thickness of a few millimeters (a thousand times greater).

PROBLEM-SOLVING STRATEGY 35.1 Interference in Thin Films

IDENTIFY *the relevant concepts:* Problems with thin films involve interference of two waves, one reflected from the film's front surface and one reflected from the back surface. Typically you'll be asked to relate the wavelength, the film thickness, and the index of refraction of the film.

SET UP *the problem* using the following steps:

1. Make a drawing showing the geometry of the film. Identify the materials that adjoin the film; their properties determine whether one or both of the reflected waves have a half-cycle phase shift.
2. Identify the target variable.

EXECUTE *the solution* as follows:

1. Apply the rule for phase changes to each reflected wave: There is a half-cycle phase shift when $n_b > n_a$ and none when $n_b < n_a$.

2. If *neither* reflected wave undergoes a phase shift, or if *both* do, use Eqs. (35.17). If only one reflected wave undergoes a phase shift, use Eqs. (35.18).
3. Solve the resulting equation for the target variable. Use the wavelength $\lambda = \lambda_0/n$ of light *in the film* in your calculations, where n is the index of refraction of the film. (For air, $n = 1.000$ to four-figure precision.)
4. If you are asked about the wave that is transmitted through the film, remember that minimum intensity in the reflected wave corresponds to maximum *transmitted* intensity, and vice versa.

EVALUATE *your answer:* Interpret your results by examining what would happen if the wavelength were changed or if the film had a different thickness.

EXAMPLE 35.4 Thin-film interference I

WITH VARIATION PROBLEMS

Suppose the two glass plates in Fig. 35.12 are two microscope slides 10.0 cm long. At one end they are in contact; at the other end they are separated by a piece of paper 0.0200 mm thick. What is the spacing of the interference fringes seen by reflection? Is the fringe at the line of contact bright or dark? Assume monochromatic light with a wavelength in air of $\lambda = \lambda_0 = 500$ nm.

IDENTIFY and SET UP Figure 35.15 depicts the situation. We'll consider only interference between the light reflected from the upper and lower surfaces of the air wedge between the microscope slides. [The top slide has a relatively great thickness, about 1 mm, so we can ignore interference between the light reflected from its upper and lower surfaces (see Fig. 35.14b).] Light travels more slowly in the glass of the slides than it does in air. Hence the wave reflected from the upper surface of the air wedge has no phase shift (see Fig. 35.13a), while the wave reflected from the lower surface has a half-cycle phase shift (see Fig. 35.13c).

EXECUTE Since only one of the reflected waves undergoes a phase shift, the condition for *destructive* interference (a dark fringe) is Eq. (35.18b):

$$2t = m\lambda_0 \quad (m = 0, 1, 2, \dots)$$

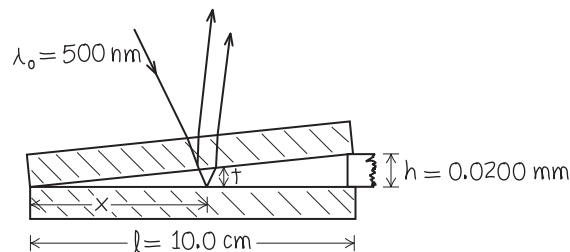
From similar triangles in Fig. 35.15 the thickness t of the air wedge at each point is proportional to the distance x from the line of contact:

$$\frac{t}{x} = \frac{h}{l}$$

Combining this with Eq. (35.18b), we find

$$\begin{aligned} \frac{2xh}{l} &= m\lambda_0 \\ x &= m \frac{l\lambda_0}{2h} = m \frac{(0.100 \text{ m})(500 \times 10^{-9} \text{ m})}{(2)(0.0200 \times 10^{-3} \text{ m})} = m(1.25 \text{ mm}) \end{aligned}$$

Figure 35.15 Our sketch for this problem.



Successive dark fringes, corresponding to $m = 1, 2, 3, \dots$, are spaced 1.25 mm apart. Substituting $m = 0$ into this equation gives $x = 0$, which is where the two slides touch (at the left-hand side of Fig. 35.15). Hence there is a dark fringe at the line of contact.

EVALUATE Our result shows that the fringe spacing is proportional to the wavelength of the light used; the fringes would be farther apart with red light (larger λ_0) than with blue light (smaller λ_0). If we use white light, the reflected light at any point is a mixture of wavelengths for which constructive interference occurs; the wavelengths that interfere destructively are weak or absent in the reflected light. (This same effect explains the colors seen when a soap bubble is illuminated by white light, as in Fig. 35.11b.)

KEYCONCEPT In thin-film interference, light waves reflected from the front surface of a thin film interfere with light waves reflected from the back surface. The phase difference between these waves depends on the thickness of the film, the wavelength of the light in the film, and whether the light undergoes a phase change on reflecting from each surface.

EXAMPLE 35.5 Thin-film interference II**WITH VARIATION PROBLEMS**

Suppose the glass plates of Example 35.4 have $n = 1.52$ and the space between plates contains water ($n = 1.33$) instead of air. What happens now?

IDENTIFY and SET UP The index of refraction of the water wedge is still less than that of the glass on either side of it, so the phase shifts are the same as in Example 35.4. Once again we use Eq. (35.18b) to find the positions of the dark fringes; the only difference is that the wavelength λ in this equation is now the wavelength in water instead of in air.

EXECUTE In the film of water ($n = 1.33$), the wavelength is $\lambda = \lambda_0/n = (500 \text{ nm})/(1.33) = 376 \text{ nm}$. When we replace λ_0 by λ in the expression from Example 35.4 for the position x of the m th dark fringe, we find that the fringe spacing is reduced by the same factor of

1.33 and is equal to 0.940 mm. There is still a dark fringe at the line of contact.

EVALUATE Can you see that to obtain the same fringe spacing as in Example 35.4, the dimension h in Fig. 35.15 would have to be reduced to $(0.0200 \text{ mm})/1.33 = 0.0150 \text{ mm}$? This shows that what matters in thin-film interference is the *ratio* t/λ between film thickness and wavelength. [Consider Eqs. (35.17) and (35.18).]

KEYCONCEPT The wavelength of interest in thin-film interference is the wavelength of light in the material of which the film is composed. This equals the wavelength in vacuum divided by the index of refraction of the material.

EXAMPLE 35.6 Thin-film interference III**WITH VARIATION PROBLEMS**

Suppose the upper of the two plates of Example 35.4 is a plastic with $n = 1.40$, the wedge is filled with a silicone grease with $n = 1.50$, and the bottom plate is a dense flint glass with $n = 1.60$. What happens now?

IDENTIFY and SET UP The geometry is again the same as shown in Fig. 35.15, but now half-cycle phase shifts occur at *both* surfaces of the grease wedge (see Fig. 35.13c). Hence there is no *relative* phase shift and we must use Eq. (35.17b) to find the positions of the dark fringes.

EXECUTE The value of λ to use in Eq. (35.17b) is the wavelength in the silicone grease, $\lambda = \lambda_0/n = (500 \text{ nm})/1.50 = 333 \text{ nm}$. You can readily show that the fringe spacing is 0.833 mm. Note that the two reflected waves from the line of contact are in phase (they both undergo the same phase shift), so the line of contact is at a *bright* fringe.

EVALUATE What would happen if you carefully removed the upper microscope slide so that the grease wedge retained its shape? There would still be half-cycle phase changes at the upper and lower surfaces of the wedge, so the pattern of fringes would be the same as with the upper slide present.

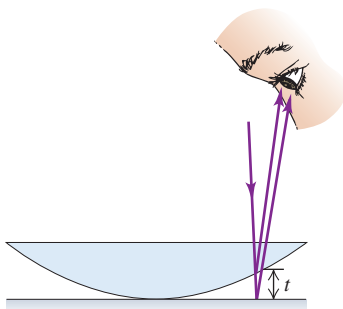
KEYCONCEPT When a light wave in one medium reflects at the surface of a second medium, it undergoes a half-cycle phase shift only if the speed of light is slower in the second medium than in the first; otherwise, there is no phase shift. If both reflected waves in thin-film interference undergo the same phase shift on reflection, there is no *relative* phase shift.

Newton's Rings

Figure 35.16a shows the convex surface of a lens in contact with a plane glass plate. A thin film of air is formed between the two surfaces. When you view the setup with monochromatic light, you see circular interference fringes (Fig. 35.16b). These were studied by Newton and are called **Newton's rings**.

Figure 35.16 (a) Air film between a convex lens and a plane surface. The thickness of the film t increases from zero as we move out from the center, giving (b) a series of alternating dark and bright rings for monochromatic light.

(a) A convex lens in contact with a glass plane



(b) Newton's rings: circular interference fringes

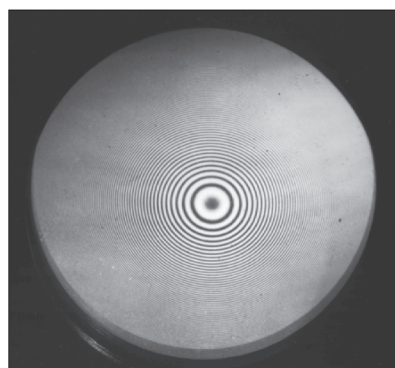


Figure 35.17 The surface of a telescope objective lens under inspection during manufacture.

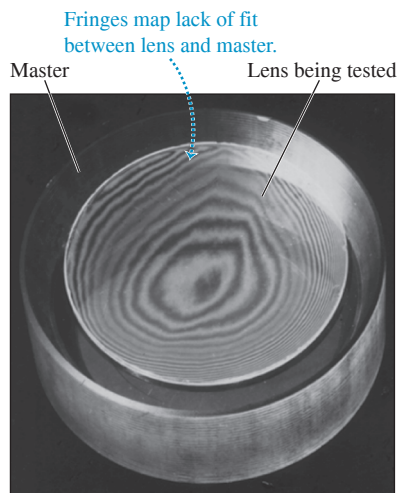
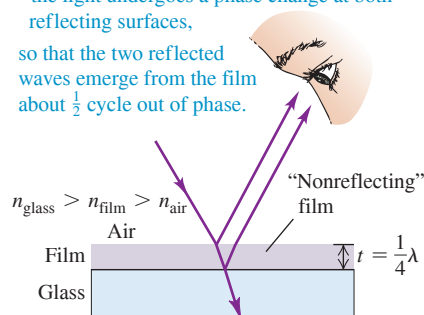


Figure 35.18 A nonreflective coating has an index of refraction intermediate between those of glass and air.

Destructive interference occurs when

- the film is about $\frac{1}{4}\lambda$ thick and
- the light undergoes a phase change at both reflecting surfaces,

so that the two reflected waves emerge from the film about $\frac{1}{2}$ cycle out of phase.



We can use interference fringes to compare the surfaces of two optical parts by placing the two in contact and observing the interference fringes. **Figure 35.17** is a photograph made during the grinding of a telescope objective lens. The lower, larger-diameter, thicker disk is the correctly shaped master, and the smaller, upper disk is the lens under test. The “contour lines” are Newton’s interference fringes; each one indicates an additional distance between the specimen and the master of one half-wavelength. At 10 lines from the center spot the distance between the two surfaces is five wavelengths, or about 0.003 mm. This isn’t very good; high-quality lenses are routinely ground with a precision of less than one wavelength. The surface of the primary mirror of the Hubble Space Telescope was ground to a precision of better than $\frac{1}{50}$ wavelength. Unfortunately, it was ground to incorrect specifications, creating one of the most precise errors in the history of optical technology (see Section 34.2).

Nonreflective and Reflective Coatings

Nonreflective coatings for lens surfaces make use of thin-film interference. A thin layer or film of hard transparent material with an index of refraction smaller than that of the glass is deposited on the lens surface, as in **Fig. 35.18**. Light is reflected from both surfaces of the layer. In both reflections the light is reflected from a medium of greater index than that in which it is traveling, so the same phase change occurs in both reflections. If the film thickness is a quarter (one-fourth) of the wavelength *in the film* (assuming normal incidence), the total path difference is a half-wavelength. Light reflected from the first surface is then a half-cycle out of phase with light reflected from the second, and there is destructive interference.

The thickness of the nonreflective coating can be a quarter-wavelength for only one particular wavelength. This is usually chosen in the central yellow-green portion of the spectrum ($\lambda = 550$ nm), where the eye is most sensitive. Then there is somewhat more reflection at both longer (red) and shorter (blue) wavelengths, and the reflected light has a purple hue. The overall reflection from a lens or prism surface can be reduced in this way from 4–5% to less than 1%. This also increases the net amount of light that is *transmitted* through the lens, since light that is not reflected will be transmitted. The same principle is used to minimize reflection from silicon photovoltaic solar cells ($n = 3.5$) by use of a thin surface layer of silicon monoxide (SiO_2 , $n = 1.45$); this helps to increase the amount of light that actually reaches the solar cells.

If a quarter-wavelength thickness of a material with an index of refraction *greater* than that of glass is deposited on glass, then the reflectivity is *increased*, and the deposited material is called a **reflective coating**. In this case there is a half-cycle phase shift at the air–film interface but none at the film–glass interface, and reflections from the two sides of the film interfere constructively. For example, a coating with refractive index 2.5 causes 38% of the incident energy to be reflected, compared with 4% or so with no coating. By use of multiple-layer coatings, it is possible to achieve nearly 100% reflection (or, if desired, 100% transmission) for particular wavelengths. Some practical applications of reflective coatings are for color separation in television cameras and for infrared “heat reflectors” in motion-picture projectors, mirrored sunglasses, and astronauts’ visors.

EXAMPLE 35.7 A nonreflective coating

A common lens coating material is magnesium fluoride (MgF_2), with $n = 1.38$. What thickness should a nonreflective coating have for 550 nm light if it is applied to glass with $n = 1.52$?

IDENTIFY and SET UP This coating is of the sort shown in Fig. 35.18. The thickness must be one-quarter of the wavelength of this light *in the coating*.

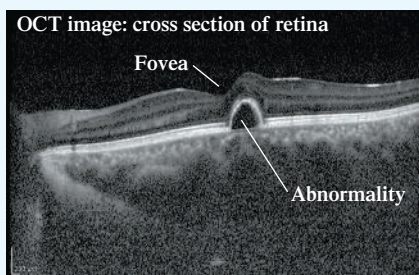
EXECUTE The wavelength in air is $\lambda_0 = 550$ nm, so its wavelength in the MgF_2 coating is $\lambda = \lambda_0/n = (550 \text{ nm})/1.38 = 400$ nm. The coating thickness should be one-quarter of this, or $\lambda/4 = 100$ nm.

EVALUATE This is a very thin film, no more than a few hundred molecules thick. Note that this coating is *reflective* for light whose wavelength is *twice* the coating thickness; light of that wavelength reflected

from the coating’s lower surface travels one wavelength farther than light reflected from the upper surface, so the two waves are in phase and interfere constructively. This occurs for light with a wavelength in MgF_2 of 200 nm and a wavelength in air of $(200 \text{ nm})(1.38) = 276$ nm. This is an ultraviolet wavelength that isn’t visible to the eye (see Section 32.1), so a lens designer need not worry about such enhanced reflection.

KEYCONCEPT A thin-film coating *minimizes* reflection for certain wavelengths by causing *destructive* interference of the reflected waves. The same coating *maximizes* reflection for other wavelengths by causing *constructive* interference of the reflected waves.

BIO APPLICATION Seeing Below the Surface with Interferometry Visible light can penetrate up to half a millimeter below the surface of biological tissue. However, only a very small fraction of the light is reflected back. To capture this light and image the subsurface features, a technique called *optical coherence tomography* (OCT) is used. OCT uses the same setup as a Michelson interferometer (Fig. 35.19), but with one of the mirrors replaced by the tissue to be studied. Only the light that reflects directly off the subsurface tissue gives constructive interference, which makes it possible to isolate that reflected light and form an image. OCT is widely used by ophthalmologists to get a cross-sectional view of the subsurface structure of the retina, which makes it easier to diagnose and monitor abnormalities of the eye.



The whole apparatus in Fig. 35.19 is mounted on a very rigid frame, and the position of mirror M_2 can be adjusted with a fine, very accurate micrometer screw. If the distances L_1 and L_2 are exactly equal and the mirrors M_1 and M_2 are exactly at right angles, the virtual image of M_1 formed by reflection at the silvered surface of plate C coincides with mirror M_2 . If L_1 and L_2 are *not* exactly equal, the image of M_1 is displaced slightly from M_2 ; and if the mirrors are not exactly perpendicular, the image of M_1 makes a slight angle with M_2 . Then the mirror M_2 and the virtual image of M_1 play the same roles as the two surfaces of a wedge-shaped thin film (see Section 35.4), and light reflected from these surfaces forms the same sort of interference fringes.

Suppose the angle between mirror M_2 and the virtual image of M_1 is just large enough that five or six vertical fringes are present in the field of view. If we now move the mirror M_2 slowly either backward or forward a distance $\lambda/2$, the difference in path length between rays 1 and 2 changes by λ , and each fringe moves to the left or right a distance equal to the fringe spacing. If we observe the fringe positions through a telescope with a crosshair eyepiece and m fringes cross the crosshairs when we move the mirror a distance y , then

$$y = m \frac{\lambda}{2} \quad \text{or} \quad \lambda = \frac{2y}{m} \quad (35.19)$$

If m is several thousand, the distance y is large enough that it can be measured with good accuracy, and we can obtain an accurate value for the wavelength λ . Alternatively, if the wavelength is known, a distance y can be measured by simply counting fringes when M_2 is moved by this distance. In this way, distances that are comparable to a wavelength of light can be measured with relative ease.

The Michelson-Morley Experiment

The original application of the Michelson interferometer was to the historic **Michelson-Morley experiment**. Before the electromagnetic theory of light became established, most physicists thought that the propagation of light waves occurred in a medium called the **ether**, which was believed to permeate all space. In 1887 the American scientists Albert Michelson and Edward Morley used the Michelson interferometer in an attempt to detect the motion of the earth through the ether. Suppose the interferometer in Fig. 35.19 is moving from left to right relative to the ether. According to the ether theory, this would lead to changes in the speed of light in the portions of the path shown as horizontal lines in the figure. There would be fringe shifts relative to the positions that the fringes would have if the instrument were at rest in the ether. Then when the entire instrument was rotated 90° , the other portions of the paths would be similarly affected, giving a fringe shift in the opposite direction.

Michelson and Morley expected that the motion of the earth through the ether would cause a shift of about four-tenths of a fringe when the instrument was rotated. The shift that was actually observed was less than a hundredth of a fringe and, within the limits of experimental uncertainty, appeared to be exactly zero. Despite its orbital motion around the sun, the earth appeared to be *at rest* relative to the ether. This negative result baffled physicists until 1905, when Albert Einstein developed the special theory of relativity (which we'll study in detail in Chapter 37). Einstein postulated that the speed of a light wave in vacuum has the same magnitude c relative to *all* inertial reference frames, no matter what their velocity may be relative to each other. The presumed ether then plays no role, and the concept of an ether has been abandoned.

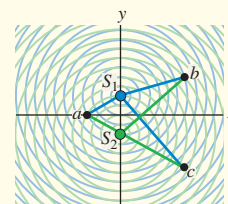
TEST YOUR UNDERSTANDING OF SECTION 35.5 You are observing the pattern of fringes in a Michelson interferometer like that shown in Fig. 35.19. If you change the index of refraction (but not the thickness) of the compensator plate, will the pattern change?

ANSWER

Yes Changing the index of refraction changes the wavelength of the light inside the compensator plate, and so changes the number of wavelengths within the thickness of the plate. Hence this has the same effect as changing the distance L_1 from the beam splitter to mirror M_1 , which would change the interference pattern.

CHAPTER 35 SUMMARY

Interference and coherent sources: Monochromatic light is light with a single frequency. Coherence is a definite, unchanging phase relationship between two waves. The overlap of waves from two coherent sources of monochromatic light forms an interference pattern. The principle of superposition states that the total wave disturbance at any point is the sum of the disturbances from the separate waves.



Two-source interference of light: When two sources are in phase, constructive interference occurs where the difference in path length from the two sources is zero or an integer number of wavelengths; destructive interference occurs where the path difference is a half-integer number of wavelengths. If two sources separated by a distance d are both very far from a point P , and the line from the sources to P makes an angle θ with the line perpendicular to the line of the sources, then the condition for constructive interference at P is Eq. (35.4). The condition for destructive interference is Eq. (35.5). When θ is very small, the position y_m of the m th bright fringe on a screen located a distance R from the sources is given by Eq. (35.6). (See Examples 35.1 and 35.2.)

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.4)$$

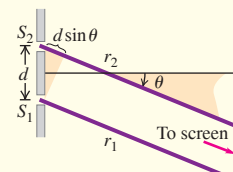
(constructive interference)

$$d \sin \theta = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.5)$$

(destructive interference)

$$y_m = R \frac{m\lambda}{d} \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.6)$$

(bright fringes)

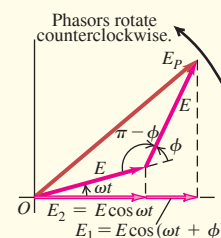


Intensity in interference patterns: When two sinusoidal waves with equal amplitude E and phase difference ϕ are superimposed, the resultant amplitude E_P and intensity I are given by Eqs. (35.7) and (35.10), respectively. If the two sources emit in phase, the phase difference ϕ at a point P (located a distance r_1 from source 1 and a distance r_2 from source 2) is directly proportional to the path difference $r_2 - r_1$. (See Example 35.3.)

$$E_P = 2E \left| \cos \frac{\phi}{2} \right| \quad (35.7)$$

$$I = I_0 \cos^2 \frac{\phi}{2} \quad (35.10)$$

$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1) \quad (35.11)$$



Interference in thin films: When light is reflected from both sides of a thin film of thickness t and no phase shift occurs at either surface, constructive interference between the reflected waves occurs when $2t$ is equal to an integer number of wavelengths. If a half-cycle phase shift occurs at one surface, this is the condition for destructive interference. A half-cycle phase shift occurs during reflection whenever the index of refraction in the second material is greater than that in the first. (See Examples 35.4–35.7.)

$$2t = m\lambda \quad (m = 0, 1, 2, \dots) \quad (35.17a)$$

(constructive reflection from thin film, no relative phase shift)

$$2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (35.17b)$$

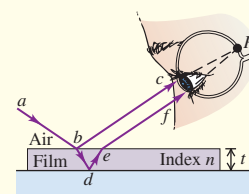
(destructive reflection from thin film, no relative phase shift)

$$2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (35.18a)$$

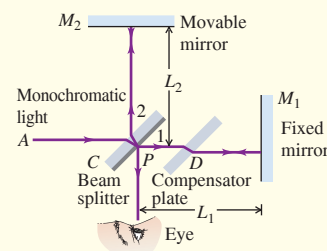
(constructive reflection from thin film, half-cycle phase shift)

$$2t = m\lambda \quad (m = 0, 1, 2, \dots) \quad (35.18b)$$

(destructive reflection from thin film, half-cycle phase shift)



Michelson interferometer: The Michelson interferometer uses a monochromatic light source and can be used for high-precision measurements of wavelengths. Its original purpose was to detect motion of the earth relative to a hypothetical ether, the supposed medium for electromagnetic waves. The ether has never been detected, and the concept has been abandoned; the speed of light is the same relative to all observers. This is part of the foundation of the special theory of relativity.





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review Examples **EXAMPLES 35.1 and 35.2** (Section 35.2) before attempting these problems.

VP35.2.1 In a two-slit interference experiment like that shown in Fig. 35.7, the slits are 0.180 mm apart and the screen is 1.50 m from the slits. The $m = 2$ bright fringe is 11.4 mm from the central bright fringe. Find (a) the wavelength of the light and (b) the distance from the central bright fringe to the $m = -3$ bright fringe.

VP35.2.2 Two identical vertical radio antennas are 0.810 km apart. Both emit in phase at a frequency of $1.05 \text{ MHz} = 1.05 \times 10^6 \text{ Hz}$. Find the angles in the range from 0° and 90° at which the intensity measured several kilometers away from these antennas is (a) maximum and (b) minimum.

VP35.2.3 You shine red laser light of wavelength 685 nm on two closely spaced slits. This creates a pattern on a screen that is 2.10 m from the double slit. The $m = 3$ bright fringe in this pattern is 9.15 mm from the central bright fringe. (a) What is the distance between the slits? (b) What would be the distance between the central and $m = 3$ bright fringes if you used green laser light of wavelength 515 nm?

VP35.2.4 In a two-slit interference experiment the distance between the slits is 0.370 mm and the angle of the line from the slits to the $m = 4$ bright region on the screen is 0.407° . Find (a) the wavelength of the light used and (b) the angle of the line from the slits to the $m = 2$ dark region on the screen.

Be sure to review **EXAMPLE 35.3** (Section 35.3) before attempting these problems.

VP35.3.1 Two identical radio antennas are 13.0 m apart and transmit in phase at frequency 59.3 MHz. At a distance of 0.850 km from the midpoint between the antennas in the direction $\theta = 0$, the intensity is $I_0 = 0.0330 \text{ W/m}^2$. Find (a) the intensity at this same distance in the direction $\theta = 5.00^\circ$ and (b) the smallest angle for which the intensity is $I_0/2$.

VP35.3.2 A two-slit interference experiment uses laser light of wavelength 655 nm. The slits are 0.230 mm apart and 1.75 m from the screen on which the interference pattern appears. The intensity of the light at the central bright fringe is $I_0 = 0.0520 \text{ W/m}^2$. Find (a) the intensity at a point on the screen 6.50 mm from the central bright fringe and (b) the distance on the screen from the central bright fringe to the nearest point where the intensity is $I_0/4$.

VP35.3.3 You walk in a circle of radius 1.20 km around the midpoint of a pair of identical radio antennas that are 8.00 m apart. The intensity decreases continuously from 0.0540 W/m^2 in the direction $\theta = 0^\circ$ to

0.0303 W/m^2 in the direction $\theta = 6.00^\circ$. Find (a) the wavelength of the radio waves and (b) the intensity in the direction $\theta = 12.0^\circ$.

VP35.3.4 A laser of unknown wavelength illuminates a pair of slits separated by 0.110 mm, producing an interference pattern on a wall 5.00 m from the slits. A point on the wall 10.5 mm from the central bright fringe is between that fringe and the $m = 1$ bright fringe, and the intensity there is 9.00% as great as at the central bright fringe. Find (a) the wavelength of the laser and (b) the distance from the central bright fringe to the nearest point on the wall where the intensity is 30.0% as great as at the central bright fringe.

Be sure to review **EXAMPLES 35.4, 35.5, and 35.6** (Section 35.4) before attempting these problems.

VP35.6.1 Two identical microscope slides made of glass ($n = 1.50$) are stacked on top of each other. The slides are 8.00 cm long and are in contact at one end; at the other end they are separated by a piece of paper. You illuminate the slides from above with monochromatic light that has a wavelength in air of 565 nm. The spacing between the interference fringes seen by reflection is 1.30 mm. (a) Find the thickness of the piece of paper. (b) What would be the spacing between fringes if the space between the slides were filled with ethanol ($n = 1.36$) rather than air?

VP35.6.2 A thin, flat slide of glass ($n = 1.50$) lies on a table, and a second, identical glass slide is suspended 146 nm above the first one. The surfaces of the slides are parallel to each other. You illuminate the two slides from above. Find the wavelengths in air, if any, in the visible range (380 to 750 nm) for which there will be constructive reflection and destructive reflection if the space between the slides is filled with (a) air or (b) carbon tetrachloride ($n = 1.46$).

VP35.6.3 A thin, flat sheet of plastic ($n = 1.40$) lies on a table, and a second sheet of the same dimensions but made of flint glass ($n = 1.60$) is suspended 172 nm above the first sheet. The surfaces of the sheets are parallel to each other. You illuminate the two sheets from above. Find the wavelengths in air, if any, in the visible range (380 to 750 nm) for which there will be constructive reflection and destructive reflection if the space between the sheets is filled with (a) air or (b) ethanol ($n = 1.36$).

VP35.6.4 A bowl of water ($n = 1.33$) has a thin layer of grease ($n = 1.60$) floating on top of the water. When you illuminate the grease from above with monochromatic light, you get constructive reflection if the wavelength in air is 565 nm. Find (a) the smallest possible thickness of the grease layer and (b) the second smallest possible thickness.

BRIDGING PROBLEM Thin-Film Interference in an Oil Slick

An oil tanker spills a large amount of oil ($n = 1.45$) into the sea ($n = 1.33$). (a) If you look down onto the oil spill from overhead, what predominant wavelength of light do you see at a point where the oil is 380 nm thick? What color is the light? (*Hint:* See Table 32.1.) (b) In the water under the slick, what visible wavelength (as measured in air) is predominant in the transmitted light at the same place in the slick as in part (a)?

SOLUTION GUIDE

IDENTIFY and SET UP

1. The oil layer acts as a thin film, so we must consider interference between light that reflects from the top and bottom surfaces of the oil. If a wavelength is prominent in the *transmitted* light, there is destructive interference for that wavelength in the *reflected* light.

2. Choose the appropriate interference equations that relate the thickness of the oil film and the wavelength of light. Take account of the indexes of refraction of the air, oil, and water.

EXECUTE

3. For part (a), find the wavelengths for which there is constructive interference as seen from above the oil film. Which of these are in the visible spectrum?
4. For part (b), find the visible wavelength for which there is destructive interference as seen from above the film. (This will ensure that there is substantial transmitted light at the wavelength.)

EVALUATE

5. If a diver below the water's surface shines a light up at the bottom of the oil film, at what wavelengths would there be constructive interference in the light that reflects back downward?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q35.1 A two-slit interference experiment is set up, and the fringes are displayed on a screen. Then the whole apparatus is immersed in the nearest swimming pool. How does the fringe pattern change?

Q35.2 Could an experiment similar to Young's two-slit experiment be performed with sound? How might this be carried out? Does it matter that sound waves are longitudinal and electromagnetic waves are transverse? Explain.

Q35.3 Monochromatic coherent light passing through two thin slits is viewed on a distant screen. Are the bright fringes equally spaced on the screen? If so, why? If not, which ones are closest to being equally spaced?

Q35.4 In a two-slit interference pattern on a distant screen, are the bright fringes midway between the dark fringes? Is this ever a good approximation?

Q35.5 Would the headlights of a distant car form a two-source interference pattern? If so, how might it be observed? If not, why not?

Q35.6 The two sources S_1 and S_2 shown in Fig. 35.3 emit waves of the same wavelength λ and are in phase with each other. Suppose S_1 is a weaker source, so that the waves emitted by S_1 have half the amplitude of the waves emitted by S_2 . How would this affect the positions of the antinodal lines and nodal lines? Would there be total reinforcement at points on the antinodal curves? Would there be total cancellation at points on the nodal curves? Explain your answers.

Q35.7 Could the Young two-slit interference experiment be performed with gamma rays? If not, why not? If so, discuss differences in the experimental design compared to the experiment with visible light.

Q35.8 Coherent red light illuminates two narrow slits that are 25 cm apart. Will a two-slit interference pattern be observed when the light from the slits falls on a screen? Explain.

Q35.9 Coherent light with wavelength λ falls on two narrow slits separated by a distance d . If d is less than some minimum value, no dark fringes are observed. Explain. In terms of λ , what is this minimum value of d ?

Q35.10 A fellow student, who values memorizing equations above understanding them, combines Eqs. (35.4) and (35.13) to "prove" that ϕ can *only* equal $2\pi m$. How would you explain to this student that ϕ can have values other than $2\pi m$?

Q35.11 If the monochromatic light shown in Fig. 35.5a were replaced by white light, would a two-slit interference pattern be seen on the screen? Explain.

Q35.12 In using the superposition principle to calculate intensities in interference patterns, could you add the intensities of the waves instead of their amplitudes? Explain.

Q35.13 A glass windowpane with a thin film of water on it reflects less than when it is perfectly dry. Why?

Q35.14 A *very* thin soap film ($n = 1.33$), whose thickness is much less than a wavelength of visible light, looks black; it appears to reflect no light at all. Why? By contrast, an equally thin layer of soapy water ($n = 1.33$) on glass ($n = 1.50$) appears quite shiny. Why is there a difference?

Q35.15 Interference can occur in thin films. Why is it important that the films be *thin*? Why don't you get these effects with a relatively *thick* film? Where should you put the dividing line between "thin" and "thick"? Explain your reasoning.

Q35.16 If we shine white light on an air wedge like that shown in Fig. 35.12, the colors that are weak in the light *reflected* from any point along the wedge are strong in the light *transmitted* through the wedge. Explain why this should be so.

Q35.17 Monochromatic light is directed at normal incidence on a thin film. There is destructive interference for the reflected light, so the intensity of the reflected light is very low. What happened to the energy of the incident light?

Q35.18 When a thin oil film spreads out on a puddle of water, the thinnest part of the film looks dark in the resulting interference pattern. What does this tell you about the relative magnitudes of the refractive indexes of oil and water?

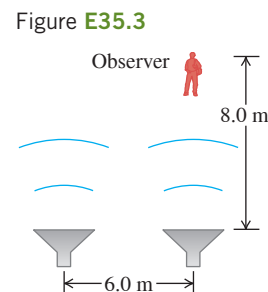
EXERCISES

Section 35.1 Interference and Coherent Sources

35.1 •• A radio transmitting station operating at a frequency of 120 MHz has two identical antennas that radiate in phase. Antenna B is 9.00 m to the right of antenna A . Consider point P between the antennas and along the line connecting them, a horizontal distance x to the right of antenna A . For what values of x will constructive interference occur at point P ?

35.2 • Radio Interference. Two radio antennas A and B radiate in phase. Antenna B is 120 m to the right of antenna A . Consider point Q along the extension of the line connecting the antennas, a horizontal distance of 40 m to the right of antenna B . The frequency, and hence the wavelength, of the emitted waves can be varied. (a) What is the longest wavelength for which there will be destructive interference at point Q ? (b) What is the longest wavelength for which there will be constructive interference at point Q ?

35.3 • Two speakers, emitting identical sound waves of wavelength 2.0 m in phase with each other, and an observer are located as shown in **Fig. E35.3**. (a) At the observer's location, what is the path difference for waves from the two speakers? (b) Will the sound waves interfere constructively or destructively at the observer's location—or something in between constructive and destructive? (c) Suppose the observer now increases her distance from the closest speaker to 17.0 m, staying directly in front of the same speaker as initially. Answer the questions of parts (a) and (b) for this new situation.



35.4 • Two light sources can be adjusted to emit monochromatic light of any visible wavelength. The two sources are coherent, $2.04 \mu\text{m}$ apart, and in line with an observer, so that one source is $2.04 \mu\text{m}$ farther from the observer than the other. (a) For what visible wavelengths (380 to 750 nm) will the observer see the brightest light, owing to constructive interference? (b) How would your answers to part (a) be affected if the two sources were not in line with the observer, but were still arranged so that one source is $2.04 \mu\text{m}$ farther away from the observer than the other? (c) For what visible wavelengths will there be *destructive* interference at the location of the observer?

35.5 •• Antenna B is 40.0 m to the right of antenna A . The two antennas emit electromagnetic waves that are in phase and have wavelength 7.00 m. (a) At how many points along the line connecting A and B is the interference constructive? (b) What is the smallest distance to the right of antenna A for which is there a point of constructive interference?

Section 35.2 Two-Source Interference of Light

35.6 •• Coherent light of wavelength 500 nm is incident on two very narrow and closely spaced slits. The interference pattern is observed on a very tall screen that is 2.00 m from the slits. Near the center of the screen the separation between two adjacent interference maxima is 3.53 cm. What is the distance on the screen between the $m = 49$ and $m = 50$ maxima?

35.7 • Young's experiment is performed with light from excited helium atoms ($\lambda = 502$ nm). Fringes are measured carefully on a screen 1.20 m away from the double slit, and the center of the 20th fringe (not counting the central bright fringe) is found to be 10.6 mm from the center of the central bright fringe. What is the separation of the two slits?

35.8 •• Coherent light with wavelength 450 nm falls on a pair of slits. On a screen 1.80 m away, the distance between dark fringes is 3.90 mm. What is the slit separation?

35.9 •• Two slits spaced 0.450 mm apart are placed 75.0 cm from a screen. What is the distance between the second and third dark lines of the interference pattern on the screen when the slits are illuminated with coherent light with a wavelength of 500 nm?

35.10 •• Coherent light of frequency f travels in air and is incident on two narrow slits. The interference pattern is observed on a distant screen that is directly opposite the slits. The frequency of light f can be varied. For $f = 5.60 \times 10^{12}$ Hz there is an interference maximum for $\theta = 60.0^\circ$. The next higher frequency for which there is an interference maximum at this angle is 7.47×10^{12} Hz. What is the separation d between the two slits?

35.11 •• Two thin parallel slits that are 0.0116 mm apart are illuminated by a laser beam of wavelength 585 nm. (a) On a very large distant screen, what is the *total* number of bright fringes (those indicating complete constructive interference), including the central fringe and those on both sides of it? Solve this problem without calculating all the angles! (*Hint:* What is the largest that $\sin \theta$ can be? What does this tell you is the largest value of m ?) (b) At what angle, relative to the original direction of the beam, will the fringe that is most distant from the central bright fringe occur?

35.12 • Coherent light with wavelength 400 nm passes through two very narrow slits that are separated by 0.200 mm, and the interference pattern is observed on a screen 4.00 m from the slits. (a) What is the width (in mm) of the central interference maximum? (b) What is the width of the first-order bright fringe?

35.13 •• Two very narrow slits are spaced $1.80 \mu\text{m}$ apart and are placed 35.0 cm from a screen. What is the distance between the first and second dark lines of the interference pattern when the slits are illuminated with coherent light with $\lambda = 550$ nm? (*Hint:* The angle θ in Eq. (35.5) is *not* small.)

35.14 •• Coherent light that contains two wavelengths, 660 nm (red) and 470 nm (blue), passes through two narrow slits that are separated by 0.300 mm. Their interference pattern is observed on a screen 4.00 m from the slits. What is the distance on the screen between the first-order bright fringes for the two wavelengths?

35.15 •• Coherent light with wavelength 600 nm passes through two very narrow slits and the interference pattern is observed on a screen 3.00 m from the slits. The first-order bright fringe is at 4.84 mm from the center of the central bright fringe. For what wavelength of light will the first-order dark fringe be observed at this same point on the screen?

35.16 •• Coherent light of frequency 6.32×10^{14} Hz passes through two thin slits and falls on a screen 85.0 cm away. You observe that the third bright fringe occurs at ± 3.11 cm on either side of the central bright fringe. (a) How far apart are the two slits? (b) At what distance from the central bright fringe will the third dark fringe occur?

Section 35.3 Intensity in Interference Patterns

35.17 •• In a two-slit interference pattern, the intensity at the peak of the central maximum is I_0 . (a) At a point in the pattern where the phase difference between the waves from the two slits is 60.0° , what is the intensity? (b) What is the path difference for 480 nm light from the two slits at a point where the phase difference is 60.0° ?

35.18 • Two slits spaced 0.260 mm apart are 0.900 m from a screen and illuminated by coherent light of wavelength 660 nm. The intensity at the center of the central maximum ($\theta = 0^\circ$) is I_0 . What is the distance on the screen from the center of the central maximum (a) to the first minimum; (b) to the point where the intensity has fallen to $I_0/2$?

35.19 • Coherent light with wavelength 500 nm passes through narrow slits separated by 0.340 mm. At a distance from the slits large compared to their separation, what is the phase difference (in radians) in the light from the two slits at an angle of 23.0° from the centerline?

35.20 •• Two slits spaced 0.0720 mm apart are 0.800 m from a screen. Coherent light of wavelength λ passes through the two slits. In their interference pattern on the screen, the distance from the center of the central maximum to the first minimum is 3.00 mm. If the intensity at the peak of the central maximum is 0.0600 W/m^2 , what is the intensity at points on the screen that are (a) 2.00 mm and (b) 1.50 mm from the center of the central maximum?

35.21 • Consider two antennas separated by 9.00 m that radiate in phase at 120 MHz, as described in Exercise 35.1. A receiver placed 150 m from both antennas measures an intensity I_0 . The receiver is moved so that it is 1.8 m closer to one antenna than to the other. (a) What is the phase difference ϕ between the two radio waves produced by this path difference? (b) In terms of I_0 , what is the intensity measured by the receiver at its new position?

Section 35.4 Interference in Thin Films

35.22 •• Two identical horizontal sheets of glass have a thin film of air of thickness t between them. The glass has refractive index 1.40. The thickness t of the air layer can be varied. Light with wavelength λ in air is at normal incidence onto the top of the air film. There is constructive interference between the light reflected at the top and bottom surfaces of the air film when its thickness is 650 nm. For the same wavelength of light the next larger thickness for which there is constructive interference is 910 nm. (a) What is the wavelength λ of the light when it is traveling in air? (b) What is the smallest thickness t of the air film for which there is constructive interference for this wavelength of light?

35.23 • What is the thinnest film of a coating with $n = 1.42$ on glass ($n = 1.52$) for which destructive interference of the red component (650 nm) of an incident white light beam in air can take place by reflection?

35.24 •• Nonglare Glass. When viewing a piece of art that is behind glass, one often is affected by the light that is reflected off the front of the glass (called *glare*), which can make it difficult to see the art clearly. One solution is to coat the outer surface of the glass with a film to cancel part of the glare. (a) If the glass has a refractive index of 1.62 and you use TiO_2 , which has an index of refraction of 2.62, as the coating, what is the minimum film thickness that will cancel light of wavelength 505 nm? (b) If this coating is too thin to stand up to wear, what other thickness would also work? Find only the three thinnest ones.

35.25 •• A uniform film of TiO_2 , 1036 nm thick and having index of refraction 2.62, is spread uniformly over the surface of crown glass of refractive index 1.52. Light of wavelength 520.0 nm falls at normal incidence onto the film from air. You want to increase the thickness of this film so that the reflected light cancels. (a) What is the *minimum* thickness of TiO_2 that you must *add* so the reflected light cancels as desired? (b) After you make the adjustment in part (a), what is the path difference between the light reflected off the top of the film and the light that cancels it after traveling through the film? Express your answer in (i) nanometers and (ii) wavelengths of the light in the TiO_2 film.

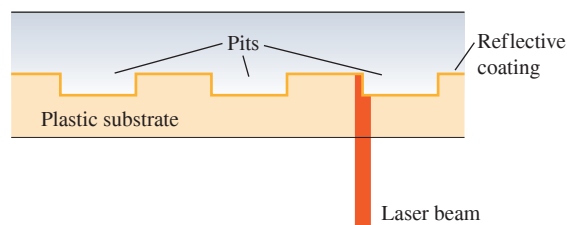
35.26 • A plastic film with index of refraction 1.70 is applied to the surface of a car window to increase the reflectivity and thus to keep the car's interior cooler. The window glass has index of refraction 1.52. (a) What minimum thickness is required if light of wavelength 550 nm in air reflected from the two sides of the film is to interfere constructively? (b) Coatings as thin as that calculated in part (a) are difficult to manufacture and install. What is the next greater thickness for which constructive interference will also occur?

35.27 • The walls of a soap bubble have about the same index of refraction as that of plain water, $n = 1.33$. There is air both inside and outside the bubble. (a) What wavelength (in air) of visible light is most strongly reflected from a point on a soap bubble where its wall is 290 nm thick? To what color does this correspond (see Fig. 32.4 and Table 32.1)? (b) Repeat part (a) for a wall thickness of 340 nm.

35.28 •• A researcher measures the thickness of a layer of benzene ($n = 1.50$) floating on water by shining monochromatic light onto the film and varying the wavelength of the light. She finds that light of wavelength 575 nm is reflected most strongly from the film. What does she calculate for the minimum thickness of the film?

35.29 •• Compact Disc Player. A compact disc (CD) is read from the bottom by a semiconductor laser with wavelength 790 nm passing through a plastic substrate of refractive index 1.8. When the beam encounters a pit, part of the beam is reflected from the pit and part from the flat region between the pits, so these two beams interfere with each other (Fig. E35.29). What must the minimum pit depth be so that the part of the beam reflected from a pit cancels the part of the beam reflected from the flat region? (It is this cancellation that allows the player to recognize the beginning and end of a pit.)

Figure E35.29



Section 35.5 The Michelson Interferometer

35.30 • Jan first uses a Michelson interferometer with the 606 nm light from a krypton-86 lamp. He displaces the movable mirror away from him, counting 818 fringes moving across a line in his field of view. Then Linda replaces the krypton lamp with filtered 502 nm light from a helium lamp and displaces the movable mirror toward her. She also counts 818 fringes, but they move across the line in her field of view opposite to the direction they moved for Jan. Assume that both Jan and Linda counted to 818 correctly. (a) What distance did each person move the mirror? (b) What is the resultant displacement of the mirror?

35.31 • How far must the mirror M_2 (see Fig. 35.19) of the Michelson interferometer be moved so that 1800 fringes of He-Ne laser light ($\lambda = 633$ nm) move across a line in the field of view?

PROBLEMS

35.32 •• The LIGO experiment, which historically detected gravitational waves for the first time in September 2015, uses a pair of highly sensitive Michelson interferometers. These have arms that are 4.00 km long and use powerful Nd:Yag lasers with 1064 nm wavelength. The beams traverse the arms both ways 280 times before recombining, which effectively lengthens the arm length to 1120 km. The devices are tuned so that the beams destructively interfere when they recombine if no gravitational wave is present. (a) The beam has a power of 100 kW, concentrated into an area of a square centimeter. Calculate the amplitude of the electric field in the beam. (b) LIGO can detect a gravitational wave that temporarily lengthens one arm by the minuscule amount of 10^{-18} m! When this happens, the beams combine with a phase difference of $\pi + \delta$. Estimate the shift δ in radians. Note that the phase difference accumulates during both traversals of each round trip. (c) Use Eq. (35.7) to estimate the sensitivity of the photodetector in terms of the minimal electric field strength needed to detect a gravitational wave.

35.33 •• One round face of a 3.25 m, solid, cylindrical plastic pipe is covered with a thin black coating that completely blocks light. The opposite face is covered with a fluorescent coating that glows when it is struck by light. Two straight, thin, parallel scratches, 0.225 mm apart, are made in the center of the black face. When laser light of wavelength 632.8 nm shines through the slits perpendicular to the black face, you find that the central bright fringe on the opposite face is 5.82 mm wide, measured between the dark fringes that border it on either side. What is the index of refraction of the plastic?

35.34 •• Newton's rings are visible when a planoconvex lens is placed on a flat glass surface. For a particular lens with an index of refraction of $n = 1.50$ and a glass plate with an index of $n = 1.80$, the diameter of the third bright ring is 0.640 mm. If water ($n = 1.33$) now fills the space between the lens and the glass plate, what is the new diameter of this ring? Assume the radius of curvature of the lens is much greater than the wavelength of the light.

35.35 • BIO Coating Eyeglass Lenses. Eyeglass lenses can be coated on the *inner* surfaces to reduce the reflection of stray light to the eye. If the lenses are medium flint glass of refractive index 1.62 and the coating is fluorite of refractive index 1.432, (a) what minimum thickness of film is needed on the lenses to cancel light of wavelength 550 nm reflected toward the eye at normal incidence? (b) Will any other wavelengths of visible light be cancelled or enhanced in the reflected light?

35.36 •• BIO Sensitive Eyes. After an eye examination, you put some eyedrops on your sensitive eyes. The cornea (the front part of the eye) has an index of refraction of 1.38, while the eyedrops have a refractive index of 1.45. After you put in the drops, your friends notice that your eyes look red, because red light of wavelength 600 nm has been reinforced in the reflected light. (a) What is the minimum thickness of the film of eyedrops on your cornea? (b) Will any other wavelengths of visible light be reinforced in the reflected light? Will any be cancelled? (c) Suppose you had contact lenses, so that the eyedrops went on them instead of on your corneas. If the refractive index of the lens material is 1.50 and the layer of eyedrops has the same thickness as in part (a), what wavelengths of visible light will be reinforced? What wavelengths will be cancelled?

35.37 •• Two flat plates of glass with parallel faces are on a table, one plate on the other. Each plate is 11.0 cm long and has a refractive index of 1.55. A very thin sheet of metal foil is inserted under the end of the upper plate to raise it slightly at that end, in a manner similar to that discussed in Example 35.4. When you view the glass plates from above with reflected white light, you observe that, at 1.15 mm from the line where the sheets are in contact, the violet light of wavelength 400.0 nm is enhanced in this reflected light, but no visible light is enhanced closer to the line of contact. (a) How far from the line of contact will green light (of wavelength 550.0 nm) and orange light (of wavelength 600.0 nm) first be enhanced? (b) How far from the line of contact will the violet, green, and orange light again be enhanced in the reflected light? (c) How thick is the metal foil holding the ends of the plates apart?

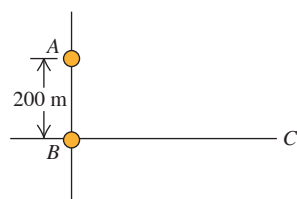
35.38 •• A typical red laser pointer has a wavelength of 650 nm. Suppose we wanted to test the wave nature of light by carefully cutting two parallel slits in a dark plastic sheet. We would then shine the laser through the slits onto a wall located 1 m beyond the sheet. (a) Determine the slit spacing needed so that the bright spots on the wall would be discernible with 1 cm spacing. (b) Is this a feasible "home experiment"? Is it possible to cut slits with that separation using typical household tools? (c) Suppose we wanted to test the wave nature of sound in a similar manner, by placing two small speakers driven in-phase near each other, separated by 40 cm, both facing a wall 2 m distant. If we used a 1.0 kHz tone, determine the distance between points along the wall that would exhibit enhanced sound. (d) Suppose we wanted the sound-enhanced points to be separated by only 1.75 m to render this experiment feasible. Estimate an audible frequency f and use it to determine a speaker separation distance d that would accomplish this. (e) Is this a feasible "home experiment"?

35.39 •• Suppose you illuminate two thin slits by monochromatic coherent light in air and find that they produce their first interference *minima* at $\pm 35.20^\circ$ on either side of the central bright spot. You then immerse these slits in a transparent liquid and illuminate them with the same light. Now you find that the first minima occur at $\pm 19.46^\circ$ instead. What is the index of refraction of this liquid?

35.40 •• CP CALC A very thin sheet of brass contains two thin parallel slits. When a laser beam shines on these slits at normal incidence and room temperature (20.0°C), the first interference dark fringes occur at $\pm 26.6^\circ$ from the original direction of the laser beam when viewed from some distance. If this sheet is now slowly heated to 135°C , by how many degrees do these dark fringes change position? Do they move closer together or farther apart? See Table 17.1 for pertinent information, and ignore any effects that might occur due to a change in the thickness of the slits. (*Hint:* Thermal expansion normally produces very small changes in length, so you can use differentials to find the change in the angle.)

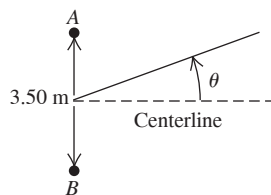
35.41 •• Two radio antennas radiating in phase are located at points A and B, 200 m apart (Fig. P35.41). The radio waves have a frequency of 5.80 MHz. A radio receiver is moved out from point B along a line perpendicular to the line connecting A and B (line BC shown in Fig. P35.41). At what distances from B will there be *destructive* interference? (*Note:* The distance of the receiver from the sources is not large in comparison to the separation of the sources, so Eq. (35.5) does not apply.)

Figure P35.41



35.42 •• Two speakers A and B are 3.50 m apart, and each one is emitting a frequency of 444 Hz. However, because of signal delays in the cables, speaker A is one-fourth of a period *ahead* of speaker B. For points far from the speakers, find all the angles relative to the centerline (Fig. P35.42) at which the sound from these speakers cancels. Include angles on *both* sides of the centerline. The speed of sound is 340 m/s.

Figure P35.42



35.43 •• CP A thin uniform film of refractive index 1.750 is placed on a sheet of glass of refractive index 1.50. At room temperature (20.0°C), this film is just thick enough for light with wavelength 582.4 nm reflected off the top of the film to be cancelled by light reflected from the top of the glass. After the glass is placed in an oven and slowly heated to 170°C , you find that the film cancels reflected light with wavelength 588.5 nm. What is the coefficient of linear expansion of the film? (Ignore any changes in the refractive index of the film due to the temperature change.)

35.44 ••• GPS Transmission. The GPS (Global Positioning System) satellites are approximately 5.18 m across and transmit two low-power signals, one of which is at 1575.42 MHz (in the UHF band). In a series of laboratory tests on the satellite, you put two 1575.42 MHz UHF transmitters at opposite ends of the satellite. These broadcast in phase uniformly in all directions. You measure the intensity at points on a circle that is several hundred meters in radius and centered on the satellite. You measure angles on this circle relative to a point that lies along the centerline of the satellite (that is, the perpendicular bisector of a line that extends from one transmitter to the other). At this point on the circle, the measured intensity is 2.00 W/m^2 . (a) At how many other angles in the range $0^\circ < \theta < 90^\circ$ is the intensity also 2.00 W/m^2 ? (b) Find the four smallest angles in the range $0^\circ < \theta < 90^\circ$ for which the intensity is 2.00 W/m^2 . (c) What is the intensity at a point on the circle at an angle of 4.65° from the centerline?

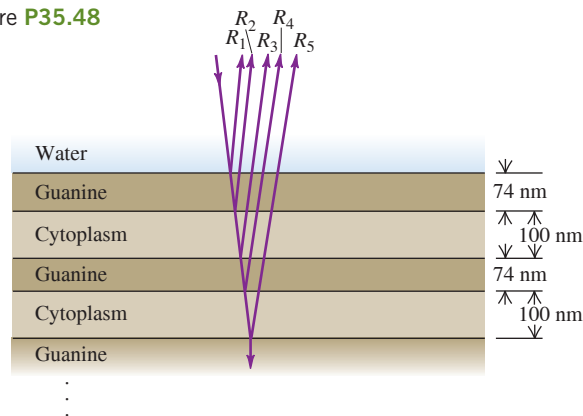
35.45 •• White light reflects at normal incidence from the top and bottom surfaces of a glass plate ($n = 1.52$). There is air above and below the plate. Constructive interference is observed for light whose wavelength in air is 477.0 nm. What is the thickness of the plate if the next longer wavelength for which there is constructive interference is 540.6 nm?

35.46 •• Laser light of wavelength 510 nm is traveling in air and shines at normal incidence onto the flat end of a transparent plastic rod that has $n = 1.30$. The end of the rod has a thin coating of a transparent material that has refractive index 1.65. What is the minimum (nonzero) thickness of the coating (a) for which there is maximum transmission of the light into the rod; (b) for which transmission into the rod is minimized?

35.47 •• Red light with wavelength 700 nm is passed through a two-slit apparatus. At the same time, monochromatic visible light with another wavelength passes through the same apparatus. As a result, most of the pattern that appears on the screen is a mixture of two colors; however, the center of the third bright fringe ($m = 3$) of the red light appears pure red, with none of the other color. What are the possible wavelengths of the second type of visible light? Do you need to know the slit spacing to answer this question? Why or why not?

35.48 •• BIO Reflective Coatings and Herring. Herring and related fish have a brilliant silvery appearance that camouflages them while they are swimming in a sunlit ocean. The silveriness is due to *platelets* attached to the surfaces of these fish. Each platelet is made up of several alternating layers of crystalline guanine ($n = 1.80$) and of cytoplasm ($n = 1.333$, the same as water), with a guanine layer on the outside in contact with the surrounding water (Fig. P35.48). In one typical platelet, the guanine layers are 74 nm thick and the cytoplasm layers are 100 nm thick. (a) For light striking the platelet surface at normal incidence, for which vacuum wavelengths of visible light will all of the reflections R_1, R_2, R_3, R_4 , and R_5 , shown in Fig. P35.48, be approximately in phase? If white light is shone on this platelet, what color will be most strongly reflected (see Fig. 32.4)? The surface of a herring has very many platelets side by side with layers of different thickness, so that *all* visible wavelengths are reflected. (b) Explain why such a “stack” of layers is more reflective than a single layer of guanine with cytoplasm underneath. (A stack of five guanine layers separated by cytoplasm layers reflects more than 80% of incident light at the wavelength for which it is “tuned.”) (c) The color that is most strongly reflected from a platelet depends on the angle at which it is viewed. Explain why this should be so. (You can see these changes in color by examining a herring from different angles. Most of the platelets on these fish are oriented in the same way, so that they are vertical when the fish is swimming.)

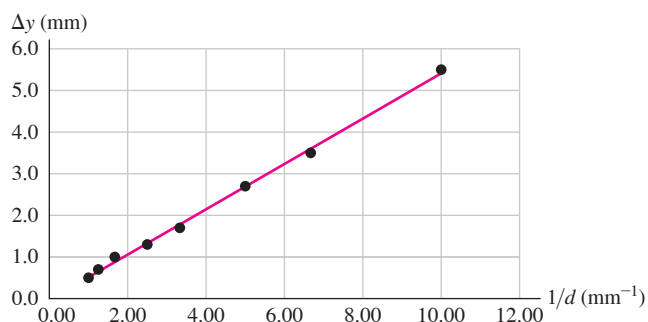
Figure P35.48



35.49 •• After a laser beam passes through two thin parallel slits, the first completely dark fringes occur at $\pm 19.0^\circ$ with the original direction of the beam, as viewed on a screen far from the slits. (a) What is the ratio of the distance between the slits to the wavelength of the light illuminating the slits? (b) What is the smallest angle, relative to the original direction of the laser beam, at which the intensity of the light is $\frac{1}{10}$ the maximum intensity on the screen?

35.50 •• DATA In your summer job at an optics company, you are asked to measure the wavelength λ of the light that is produced by a laser. To do so, you pass the laser light through two narrow slits that are separated by a distance d . You observe the interference pattern on a screen that is 0.900 m from the slits and measure the separation Δy between adjacent bright fringes in the portion of the pattern that is near the center of the screen. Using a microscope, you measure d . But both Δy and d are small and difficult to measure accurately, so you repeat the measurements for several pairs of slits, each with a different value of d . Your results are shown in **Fig. P35.50**, where you have plotted Δy versus $1/d$. The line in the graph is the best-fit straight line for the data. (a) Explain why the data points plotted this way fall close to a straight line. (b) Use **Fig. P35.50** to calculate λ .

Figure P35.50

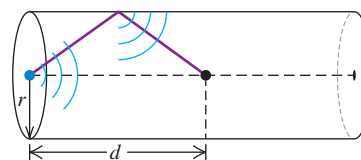


35.51 •• DATA Short-wave radio antennas A and B are connected to the same transmitter and emit coherent waves in phase and with the same frequency f . You must determine the value of f and the placement of the antennas that produce a maximum intensity through constructive interference at a receiving antenna that is located at point P , which is at the corner of your garage. First you place antenna A at a point 240.0 m due east of P . Next you place antenna B on the line that connects A and P , a distance x due east of P , where $x < 240.0$ m. Then you measure that a maximum in the total intensity from the two antennas occurs when $x = 210.0$ m, 216.0 m, and 222.0 m. You don't investigate smaller or larger values of x . (Treat the antennas as point sources.) (a) What is the frequency f of the waves that are emitted by the antennas? (b) What is the greatest value of x , with $x < 240.0$ m, for which the interference at P is destructive?

35.52 •• DATA In your research lab, a very thin, flat piece of glass with refractive index 1.40 and uniform thickness covers the opening of a chamber that holds a gas sample. The refractive indexes of the gases on either side of the glass are very close to unity. To determine the thickness of the glass, you shine coherent light of wavelength λ_0 in vacuum at normal incidence onto the surface of the glass. When $\lambda_0 = 496$ nm, constructive interference occurs for light that is reflected at the two surfaces of the glass. You find that the next shorter wavelength in vacuum for which there is constructive interference is 386 nm. (a) Use these measurements to calculate the thickness of the glass. (b) What is the longest wavelength in vacuum for which there is constructive interference for the reflected light?

35.53 •• CP An acoustic waveguide consists of a long cylindrical tube with radius r designed to channel sound waves, as shown in **Fig. P35.53**. A tone with frequency f is emitted from a small source at the center of one end of this tube. Depending on the radius of the tube and the frequency of the tone, pressure nodes can develop along the tube axis where rays reflected from the periphery constructively interfere with direct rays. (a) For sound waves with wavelength λ , what is the minimum tube radius for which at least one such node exists? (b) The tube has radius 25.0 cm and the temperature is 20°C. If the tone has frequency 2.50 kHz, how many nodes exist? (c) At what distance d are these nodes located? (d) If the tube were filled with helium rather than air, how many nodes would exist, and at what value of d ?

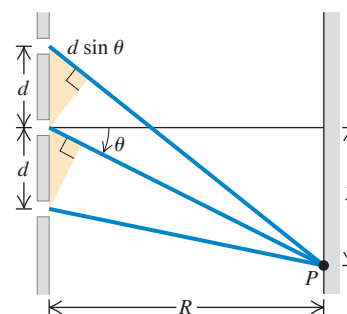
Figure P35.53



CHALLENGE PROBLEMS

35.54 ••• Three-Slit Interference. Monochromatic light with wavelength λ is incident on a screen with three narrow slits with separation d , as shown in **Fig. P35.54**. Light from the middle slit reaches point P with electric field $E \cos(\omega t)$. From the small-angle approximation, light from the upper and lower slits reaches point P with electric fields $E \cos(\omega t + \phi)$ and $E \cos(\omega t - \phi)$, respectively, where $\phi = (2\pi d \sin \theta)/\lambda$ is the phase lag and phase lead associated with the different path lengths. (a) Using either a phasor analysis similar to **Fig. 35.9** or trigonometric identities, determine the electric-field amplitude E_P associated with the net field at point P , in terms of ϕ . (b) Determine the intensity I at point P in terms of the maximum net intensity I_0 and the phase angle ϕ . (c) There are two sets of relative maxima: one with intensity I_0 when $\phi = 2\pi m$, where m is an integer, and another with a smaller intensity at other values of ϕ . What values of ϕ exhibit the "lesser" maxima? (d) What is the intensity at the lesser maxima, in terms of I_0 ? (e) What values of ϕ correspond to the dark fringes closest to the center? (f) If the incident light has wavelength 650 nm, the slits are separated by $d = 0.200$ mm, and the distance to the far screen is $R = 1.00$ m, what is the distance from the central maximum to the first lesser maximum? (g) What is the distance from the central maximum to the closest absolute maximum?

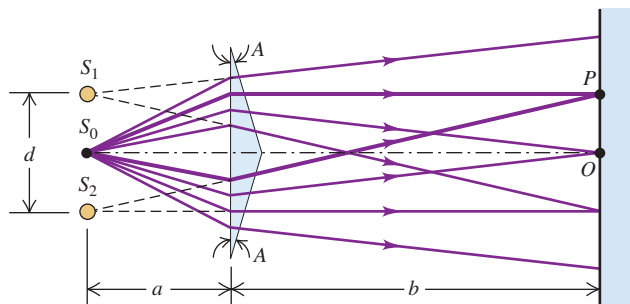
Figure P35.54



35.55 ••• CP The index of refraction of a glass rod is 1.48 at $T = 20.0^\circ\text{C}$ and varies linearly with temperature, with a coefficient of $2.50 \times 10^{-5}/^\circ\text{C}$. The coefficient of linear expansion of the glass is $5.00 \times 10^{-6}/^\circ\text{C}$. At 20.0°C the length of the rod is 3.00 cm. A Michelson interferometer has this glass rod in one arm, and the rod is being heated so that its temperature increases at a rate of $5.00^\circ\text{C}/\text{min}$. The light source has wavelength $\lambda = 589$ nm, and the rod initially is at $T = 20.0^\circ\text{C}$. How many fringes cross the field of view each minute?

35.56 ••• CP Figure P35.56 shows an interferometer known as *Fresnel's biprism*. The magnitude of the prism angle A is extremely small. (a) If S_0 is a very narrow source slit, show that the separation of the two virtual coherent sources S_1 and S_2 is given by $d = 2aA(n - 1)$, where n is the index of refraction of the material of the prism. (b) Calculate the spacing of the fringes of green light with wavelength 500 nm on a screen 2.00 m from the biprism. Take $a = 0.200$ m, $A = 3.50$ mrad, and $n = 1.50$.

Figure P35.56



MCAT-STYLE PASSAGE PROBLEMS

Interference and Sound Waves. Interference occurs with not only light waves but also all frequencies of electromagnetic waves and all other types of waves, such as sound and water waves. Suppose that your physics professor sets up two sound speakers in the front of your classroom and uses an electronic oscillator to produce sound waves of a single frequency. When she turns the oscillator on (take this to be its original setting), you and many students hear a loud tone while other students hear nothing. (The speed of sound in air is 340 m/s.)

35.57 The professor then adjusts the apparatus. The frequency that you hear does not change, but the loudness decreases. Now all of your fellow students can hear the tone. What did the professor do? (a) She turned off the oscillator. (b) She turned down the volume of the speakers. (c) She changed the phase relationship of the speakers. (d) She disconnected one speaker.

35.58 The professor returns the apparatus to the original setting. She then adjusts the speakers again. All of the students who had heard nothing originally now hear a loud tone, while you and the others who had originally heard the loud tone hear nothing. What did the professor do? (a) She turned off the oscillator. (b) She turned down the volume of the speakers. (c) She changed the phase relationship of the speakers. (d) She disconnected one speaker.

35.59 The professor again returns the apparatus to its original setting, so you again hear the original loud tone. She then slowly moves one speaker away from you until it reaches a point at which you can no longer hear the tone. If she has moved the speaker by 0.34 m (farther from you), what is the frequency of the tone? (a) 1000 Hz; (b) 2000 Hz; (c) 500 Hz; (d) 250 Hz.

35.60 The professor once again returns the apparatus to its original setting, but now she adjusts the oscillator to produce sound waves of half the original frequency. What happens? (a) The students who originally heard a loud tone again hear a loud tone, and the students who originally heard nothing still hear nothing. (b) The students who originally heard a loud tone now hear nothing, and the students who originally heard nothing now hear a loud tone. (c) Some of the students who originally heard a loud tone again hear a loud tone, but others in that group now hear nothing. (d) Among the students who originally heard nothing, some still hear nothing but others now hear a loud tone.

ANSWERS

Chapter Opening Question ?

(v) The colors appear as a result of constructive interference between light waves reflected from the upper and lower surfaces of the oil film. The wavelength of light for which the most constructive interference occurs at a point, and hence the color that appears the brightest at that point, depends on (1) the thickness of the film (which determines the path difference between light waves that reflect off the two surfaces), (2) the oil's index of refraction (which gives the wavelength of light in the oil a different value than in air), and (3) the index of refraction of the material below the oil (which determines whether the wave that reflects from the lower surface undergoes a half-cycle phase shift). (See Examples 35.4, 35.5, and 35.6 in Section 35.4.)

Key Example **✓** VARIATION Problems

VP35.2.1 (a) 684 nm (b) 17.1 mm

VP35.2.2 (a) 0° , 20.7° , 44.9° (b) 10.2° , 31.9° , 61.9°

VP35.2.3 (a) 0.472 mm (b) 6.88 mm

VP35.2.4 (a) 657 nm (b) 0.254°

VP35.3.1 (a) 0.0192 W/m^2 (b) 5.58°

VP35.3.2 (a) 0.0173 W/m^2 (b) 1.66 mm

VP35.3.3 (a) 3.63 m (b) $9.14 \times 10^{-4} \text{ W/m}^2$

VP35.3.4 (a) 573 nm (b) 8.22 mm

VP35.6.1 (a) 0.0174 mm (b) 0.956 mm

VP35.6.2 (a) constructive: 584 nm; destructive: none (b) constructive: none; destructive: 426 nm

VP35.6.3 (a) constructive: none; destructive: 688 nm (b) constructive: 468 nm; destructive: none

VP35.6.4 (a) 88.3 nm (b) 265 nm

Bridging Problem

(a) 441 nm; violet

(b) 551 nm